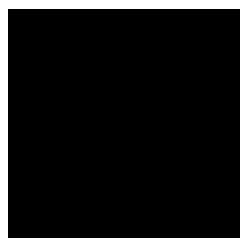


Form Z14

(February 2025)



The **ACT**[®]

2024 | 2025

In response to your request for Test Information Release materials, this booklet contains the test questions, scoring keys, and conversion tables used in determining your ACT scores. Enclosed with this booklet is a report that lists each of your answers, shows whether your answer was correct, and, if your answer was not correct, gives the correct answer.

ACT owns the test questions and responses, and you may not share them with anyone in any form.

Directions

This booklet contains tests in English, mathematics, reading, and science. These tests measure skills and abilities highly related to high school course work and success in college. **Calculators may be used on the mathematics test only.**

The questions in each test are numbered, and the suggested answers for each question are lettered. On the answer document, the rows of ovals are numbered to match the questions, and the ovals in each row are lettered to correspond to the suggested answers.

For each question, first decide which answer is best. Next, locate on the answer document the row of ovals numbered the same as the question. Then, locate the oval in that row lettered the same as your answer. Finally, fill in the oval completely. Use a soft lead pencil and make your marks heavy and black. **Do not use ink or a mechanical pencil.**

Mark only one answer to each question. If you change your mind about an answer, erase your first mark thoroughly before marking your new answer. For each question, make certain that you mark in the row of ovals with the same number as the question.

Only responses marked on your answer document will be scored. Your score on each test will be based only on the number of questions you answer correctly during the time allowed for that test. You will **not** be penalized for guessing. **It is to your advantage to answer every question even if you must guess.**

You may work on each test **only** when the testing staff tells you to do so. If you finish a test before time is called for that test, you should use the time remaining to reconsider questions you are uncertain about in that test. You may **not** look back to a test on which time has already been called, and you may **not** go ahead to another test. To do so will disqualify you from the examination.

Lay your pencil down immediately when time is called at the end of each test. You may **not** for any reason fill in or alter ovals for a test after time is called for that test. To do so will disqualify you from the examination.

Do not fold or tear the pages of your test booklet.

**DO NOT OPEN THIS BOOKLET
UNTIL TOLD TO DO SO.**

ENGLISH TEST

45 Minutes—75 Questions

DIRECTIONS: In the five passages that follow, certain words and phrases are underlined and numbered. In the right-hand column, you will find alternatives for the underlined part. In most cases, you are to choose the one that best expresses the idea, makes the statement appropriate for standard written English, or is worded most consistently with the style and tone of the passage as a whole. If you think the original version is best, choose “NO CHANGE.” In some cases, you will find in the right-hand column a question about the underlined part. You are to choose the best answer to the question.

You will also find questions about a section of the passage, or about the passage as a whole. These questions do not refer to an underlined portion of the passage, but rather are identified by a number or numbers in a box.

For each question, choose the alternative you consider best and fill in the corresponding oval on your answer document. Read each passage through once before you begin to answer the questions that accompany it. For many of the questions, you must read several sentences beyond the question to determine the answer. Be sure that you have read far enough ahead each time you choose an alternative.

PASSAGE I

Fosbury’s Victorious Flop

As Dick Fosbury watched, his teammates hurtled over the high jump bar. Raised to a height of 5’6”, which had proved to be too high, the sophomore could not clear the bar. His peers took long strides toward the bar, which rested horizontally on the notches of two

skinny posts, and launched them off the ground.

Most dove over and landed on the mat, victorious.

The high jump ruined Fosbury’s chances of becoming a star high school athlete.

1. A. NO CHANGE
B. the sophomore could not clear the bar, which had proved to be too high.
C. clearing the bar had proved to be too high for the sophomore.
D. the bar had proved too high for the sophomore to clear.
2. F. NO CHANGE
G. themselves
H. himself
J. oneself
3. A. NO CHANGE
B. over, and,
C. over and,
D. over, and
4. Which choice most effectively introduces the main focus of the essay?
F. NO CHANGE
G. Fosbury had also tried playing on his school’s basketball and football teams.
H. The high jump was one of many track-and-field events that Fosbury tried.
J. Fosbury was determined to clear that height himself.

When Fosbury joined his high school's track-and-field team in 1963, most high jumpers used⁵ the straddle technique to clear the bar. To perform the straddle, an athlete must run toward the bar at a sharp angle then launching⁶ off the inside foot (nearest the bar). Jumping into the air and stretching the outside leg

over the bar, the athlete ducks there head⁷ and tumbles over facedown, headfirst.

The straddle felt unnatural to Fosbury⁸; he decided to try the less common scissor kick technique, which requires the athlete to launch off the outside foot. Once in the air, lifting⁹ first the inside leg and then the outside leg over the bar, keeping the torso perpendicular to the ground and

allowing the athlete to land feetfirst. Whether¹⁰ Fosbury

attempted the technique¹¹ but still fell short.

The only solution was to tweak his form. Fosbury found that if he lifted his hips after launching into the air, his head, and shoulders would tilt back¹², so that he sailed over the bar headfirst and chest-up, almost parallel to the ground. The adjustment worked. In one day, Fosbury improved his personal record by six full inches.

In 1968, after perfecting his unconventional technique, dubbed the "Fosbury Flop," Fosbury qualified that same year for the¹³ Olympics in Mexico City. The audience laughed at Fosbury's form at first.

5. A. NO CHANGE
B. 1963, most high jumpers using
C. 1963. Most high jumpers using
D. 1963; most high jumpers used
6. F. NO CHANGE
G. and then launch
H. and launching
J. then launches
7. A. NO CHANGE
B. his or her heads
C. his or her head
D. their heads
8. F. NO CHANGE
G. Fosbury so
H. Fosbury,
J. Fosbury
9. A. NO CHANGE
B. the jumper lifting
C. the jumper lifts
D. DELETE the underlined portion.
10. F. NO CHANGE
G. Not only had
H. Either
J. DELETE the underlined portion.
11. The writer wants to emphasize the effort Fosbury put into executing the scissor kick technique. Which choice best accomplishes that goal?
A. NO CHANGE
B. jump after jump
C. to clear the bar
D. to jump
12. F. NO CHANGE
G. air his head, and shoulders, would tilt back,
H. air, his head and shoulders would tilt back
J. air his head and shoulders, would tilt back
13. A. NO CHANGE
B. himself for the
C. for the 1968
D. for the



But by the time he had won the gold medal and broken the Olympic record (with a height of 7'4"), Fosbury was the fan favorite and an inspiration to a new generation of high

jumpers. After his victory in Mexico City, Fosbury never again qualified for the Olympics.

14. The writer is considering deleting the underlined portion (adjusting the punctuation as needed). Should the underlined portion be kept or deleted?
- F. Kept, because it indicates how much higher Fosbury jumped than did the silver and bronze medalists.
 - G. Kept, because it shows how much Fosbury's high jump had improved as a result of the Fosbury Flop.
 - H. Deleted, because it fails to indicate whether Fosbury used the Fosbury Flop method for his record-breaking jump.
 - J. Deleted, because it interrupts the paragraph's explanation of Fosbury's high-jumping form.
15. Given that all the choices are true, which one best concludes the essay?
- A. NO CHANGE
 - B. Since 1972, every athlete who has won an Olympic medal in the high jump has done so with the Fosbury Flop.
 - C. After winning the gold, Fosbury appeared on the Johnny Carson show along with comedian Carol Burnett.
 - D. Decades later, a car company featured the Fosbury Flop in a television commercial.

PASSAGE II

Flamenco Flair

With an audience of hundreds chanting enthusiastically, flamenco dancer Mercedes Ruiz swishes her ruffled red dress over the Villamarta Theatre stage. Her footwork, proclaiming joy as she kicks up a heel and exuding passion as she stamps across the floor, creates a complex pattern. Meanwhile, guitarists strum flurries of bright notes then musicians seated nearby begin punctuated clapping to the music. Captivated by the show, cheering is prompted as Ruiz spins.

16. F. NO CHANGE
G. dancer, Mercedes Ruiz,
H. dancer, Mercedes Ruiz
J. dancer Mercedes Ruiz,
17. A. NO CHANGE
B. notes, other
C. notes, and
D. notes,
18. F. NO CHANGE
G. each of Ruiz's spins prompts cheering from the audience.
H. Ruiz's spins prompt cheering from the audience.
J. the audience cheers with each of Ruiz's spins.

She offers beguiling smiles, seamlessly transitions between

19

movements that are each interesting and unique. Together,

20

the performers drive the rhythm to a fast, feverish

celebration of one of Spain's most treasured art

forms: flamenco.

Ruiz's lively performance took place in

2011 at Spain's Festival de Jerez, an annual

celebration of flamenco. Originally, the art

21

form first blossomed between Spain's Andalusian

22

population in the eighteenth century. Flamenco

has inspired a wide variety of flamenco dance costumes.

23

The form, in its many interpretations, ranges

24

in the light, humorous *chico* style to the serious *jondo*

25

style. 26 Despite the genre's varied influences and styles, three basic elements combine to create flamenco: music, dance, and clapping. Typically, a flamenco guitarist provides a musical foundation through the use of strumming, improvised plucking, and steady rhythm.

19. A. NO CHANGE

B. smiles, then seamlessly transitioning

C. smiles and seamlessly transitions

D. smiles. Seamlessly transitions

20. At this point, the writer wants to emphasize the range of movements Ruiz makes while dancing. Which choice best accomplishes that goal?

F. NO CHANGE

G. sharp and forceful and those that are flowing and graceful.

H. practiced and display her expertise.

J. all equally beautiful.

21. A. NO CHANGE

B. In the beginning, the

C. Initially, the

D. The

22. F. NO CHANGE

G. among

H. into

J. by

23. Given that all the choices are accurate, which one most clearly provides information about the origins of flamenco?

A. NO CHANGE

B. is often mistaken for a dance that uses castanets, a percussion instrument.

C. can take place in both formal and informal settings.

D. draws from Arab, Indian, and Jewish influences.

24. F. NO CHANGE

G. their

H. it's

J. its'

25. A. NO CHANGE

B. from

C. for

D. of

26. At this point, the writer is considering dividing the paragraph into two. Should the writer begin or not begin a new paragraph here, and why?

F. Begin a new paragraph, because the essay shifts here to discuss the specifics of the three main elements of flamenco.

G. Begin a new paragraph, because the essay shifts here to compare the various cultural influences on flamenco.

H. DO NOT begin a new paragraph, because doing so would disrupt the explanation of the *chico* and *jondo* styles of flamenco.

J. DO NOT begin a new paragraph, because doing so would overemphasize the guitarist's role in the flamenco performance.

The dancer who conveys the mood with
27

neverending arm maneuvers and with brushing
28
and striking footsteps. Additional performers called
palmistas participate through a form of clapping called
palmas, their sounds enhancing quiet or exciting moments.
29

Adding to the intensity is *duende*, the collaborative
ambiance created among the participants. This *duende*
infuses Ruiz's energetic appearances, whether they are
formal concerts or casual events. Ruiz even led a group
of festivalgoers in a choreographed flash mob at the 2013
Festival de Jerez, inspiring visitors from across the globe
to step into the world of flamenco.

27. A. NO CHANGE
B. who, conveying
C. conveying
D. conveys
28. F. NO CHANGE
G. consecutive, calculated locomotion of the arms
H. arm moves that just go on and on
J. continuous arm movements
29. A. NO CHANGE
B. his or her
C. whose
D. our

Question 30 asks about the preceding passage as a whole.

30. Suppose the writer's primary purpose had been to offer a biographical overview of a successful dancer. Would this essay accomplish that purpose?
- F. Yes, because it describes the performances that defined Ruiz's career and made Ruiz a star in the dance world.
G. Yes, because it profiles Ruiz from her childhood dance experiences to her appearance at a 2013 flamenco festival.
H. No, because it instead focuses on describing what flamenco is, using Ruiz's festival performances as an example.
J. No, because it instead focuses on Ruiz's work with amateur flamenco artists.

PASSAGE III

Charles Cushman's Kodachrome America

[1] On September 3, 1938, he traveled up the Pacific
31

coast, hobbyist photographer Charles Cushman loaded his
32
first roll of Kodachrome color film into his 35-millimeter camera. [2] He began by photographing his shiny red Ford coupe, the newly painted crimson-orange Golden Gate Bridge in the background. [3] A hazy, pale-blue sky filled out the frame. [4] Though color film for cameras was
33
introduced in the mid-1930s, for decades afterwards most photographers continued to work in black and white.

31. A. NO CHANGE
B. he was traveling
C. while traveling
D. travel
32. F. NO CHANGE
G. photographer, Charles Cushman,
H. photographer, Charles Cushman
J. photographer Charles Cushman,
33. A. NO CHANGE
B. It's true that color
C. Still, color
D. Color



[5] But capturing street scenes, buildings, and landscapes

in vivid color became Cushman's passion. [6] From 1938³⁴

to 1969, he traveled up and down again and again, driving³⁵
over a half-million miles to take thousands of photographs

with color film. 36

A body, Cushman's documentary-style shots³⁷

vibrantly show Depression-era America's transition³⁸
to an age of industrial development and metropolitan
growth. He captures historical details in brilliant
color. A 1930s photo taken near an Indiana farm
depicts a gleaming black-and-yellow locomotive
heading one way, not a scratch visible on its surface.³⁹

34. F. NO CHANGE
G. one photographer's
H. their
J. his

35. A. NO CHANGE
B. went one way and then another over the country,
C. crisscrossed the country,
D. made passes,

36. The writer is considering adding the following sentence to this paragraph:

Color film was expensive, and many prominent photographers of the day argued that color in photographs was distracting, even silly.

If the writer were to add this sentence, it would most logically be placed after:

- F. Sentence 1.
G. Sentence 2.
H. Sentence 4.
J. Sentence 5.
37. A. NO CHANGE
B. Taken as a body of work that someone may view at once, as a singular element,
C. In its essence as a collected body of work together,
D. As a body of work,
38. F. NO CHANGE
G. is showing
H. has shown
J. shows
39. Given that all the choices are accurate, which one most strongly emphasizes that the 1930s photo is an example of Cushman capturing America during a time of transition?
A. NO CHANGE
B. while a man leads a tan horse and weathered cart in the other.
C. as a train engineer, wearing a white hat, sits in the cab.
D. almost appearing to be gaining speed after a stop.



In a 1949 picture, four young Chicago girls, one wearing a red-and-white gingham dress and then wrapped in a

40

blue shawl, line up to buy confections out of a rusty green cart. A majestic old building is shown half-demolished in a 1950 image—piles of crushed gray stone rest among cracked beams of wood darkened by rain to a rich mahogany.

Cushman never sought either to display or to be recognized for his work during his lifetime. So, meanwhile, his 14,500 color images

42

and a dozen (of his notebooks) that include annotations for each are archived and celebrated at his alma mater, Indiana University. To view Cushman's photographs is to see a long-gone era, one typically presented to us through black-and-white photography, in our now-familiar backdrop of full color. 44 An unreachable past feels eerily close when Cushman reminds us that the sky

on a clear, autumn day, has always been blue, even during the Depression.

45

40. Which choice makes clear that the clothing of two of the four girls in the photo is being described in the sentence?

- F. NO CHANGE
- G. another
- H. also
- J. DELETE the underlined portion.

41. A. NO CHANGE
B. were in line
C. was lined
D. lines

42. F. NO CHANGE
G. But in that case,
H. Yet similarly,
J. But now,

43. A. NO CHANGE
B. (and a dozen of his notebooks that include annotations for each)
C. and a dozen of his notebooks that include (annotations for each)
D. and (a dozen of his notebooks) that include annotations for each

44. At this point, the writer is considering adding the following true sentence:

“Buena Vista Inn on Buena Vista East of Haight St. San Francisco” is the annotation on a 1953 image.

Should the writer make this addition here?

- F. Yes, because it makes clear that although Cushman annotated every photograph he took, he didn't like to philosophize about his work.
- G. Yes, because it verifies that Cushman was aware of the relevance of his photographs.
- H. No, because it interrupts the discussion of the way Cushman's photographs might make a viewer feel.
- J. No, because it doesn't provide information that explains Cushman's feelings about the Buena Vista Inn.

45. A. NO CHANGE
B. clear autumn day
C. clear, autumn, day,
D. clear autumn day,

PASSAGE IV

Keys for Community

[1] Jostling with the crowds on a busy city morning, I hardly expected to hear the delicate notes of a piano étude. [2] There, a woman without sheet music or traditional concert attire ⁴⁶ played on an old piano. [3] What I thought must be coming

from a storefront's speakers but it was actually ⁴⁷ drifting from the city square. [4] Drawing closer, I read the message scrawled in crayon across the piano's keys: "Play Me, I'm Yours." ⁴⁸

From London and Sydney, to Los Angeles ⁴⁹ and Cincinnati; ⁴⁹ street pianos are filling ⁴⁹ cities like mine with music. Founded by British ⁵⁰ artist Luke Jerram, the "Play Me, I'm Yours" project aims to promote interaction among city dwellers, whom Jerram perceives as coexisting, but not communicating. Secondhand pianos—⁵¹ strategically placed in sights ⁵² with high pedestrian traffic, encourage people to stop a moment and interact with others.

46. If the writer were to delete the underlined portion, the sentence would primarily lose details that:

- F. indicate that the narrator was used to seeing this piano played by famous musicians.
- G. characterize the woman's playing as unexceptional.
- H. suggest that the woman's performance was spontaneous and informal.
- J. support the description of the piano as old and run-down.

47. A. NO CHANGE
B. speakers turned out to be
C. speakers, it was actually
D. speakers, instead it was

48. For the sake of logic and cohesion, Sentence 3 should be placed:

- F. where it is now.
- G. before Sentence 1.
- H. after Sentence 1.
- J. after Sentence 4.

49. A. NO CHANGE
B. Sydney, to Los Angeles and Cincinnati
C. Sydney to Los Angeles, and Cincinnati
D. Sydney to Los Angeles and Cincinnati,

50. F. NO CHANGE
G. a range of cities, including these but also my own,
H. cities like mine (in that they have street pianos)
J. these cities and others in between, like mine,

51. A. NO CHANGE
B. pianos,
C. pianos;
D. pianos

52. F. NO CHANGE
G. on sights
H. to sites
J. at sites

Often, Jerram has learned, a song is all the spark people
53

need to greet longtime neighbors or find new friends.
54

Street piano projects also offer a second chance to
54
pianos that might otherwise be scrapped. Given rising
costs of maintenance, making piano owners likely to throw
55
their old instruments away rather than have them repaired
for sale or donation. Street pianos, though, need not be
kept in perfect condition. A piano with a few worn keys
can still introduce children to the instrument or provide
students with practice opportunities. Easy to spot, a street
56
piano can also reveal the gifted musicians in our midst.

Harmonies wafted on the air. People drifted away
from storefronts. Lifting her fingers from the keys, letting
the last chord ring, the woman rose to leave. She was
brought to a halt by her now plentiful audience and its
57
fervent cheers—mine loudest among them.

Now, the next player took the bench, and the next, the
58
two of us chatted about music, the city, and ourselves. The
shuffling of foot traffic faded. Replacing it were the lively
strains of “Happy Birthday” and holiday tunes, Billy Joel
and Bach, all underscored by the hum of conversation.
59

53. A. NO CHANGE
B. Jerram has discovered as the project’s founder,
C. what Jerram finds is frequently that
D. as per Jerram,
54. F. NO CHANGE
G. neighbors or allowing them to
H. neighbors by which they
J. neighbors, also,
55. A. NO CHANGE
B. which have made piano owners
C. as a result, piano owners are
D. piano owners are
56. Which choice best returns the essay to the point of view presented in the first paragraph?
F. NO CHANGE
G. For listeners, like me,
H. Jerram has said that
J. Located downtown,
57. A. NO CHANGE
B. bountiful
C. spacious
D. sizeable
58. F. NO CHANGE
G. Meanwhile,
H. Then
J. As
59. Which choice best concludes the sentence and the essay by referring to Jerram’s main purpose in founding the street piano project?
A. NO CHANGE
B. showing that music is indeed important after all.
C. with people of all ages receiving a free concert.
D. performed by a number of talented musicians.



Question 60 asks about the preceding passage as a whole.

60. Suppose the writer's primary purpose had been to evaluate a particular musician's technique. Would this essay accomplish that purpose?
- F. Yes, because it provides a firsthand account of one person's response to a piano performance.
 - G. Yes, because it describes several attributes of a song performed by a street pianist.
 - H. No, because it focuses more directly on the street piano project and one person's encounter with it.
 - J. No, because it instead gives an overview of the fine arts opportunities offered in a particular city.

PASSAGE V

More Than Meets the Eye

[1]

[A] In 1963, the year Shiv Kumar, an ⁶¹astronomer studying stars of very low mass, theorized ⁶¹that the process that forms stars could also create a new class of smaller objects, distinct from both stars and planets. These “brown dwarfs,” as astronomer Jill Tarter ⁶²later termed them, raised essential ponderings ⁶³about how matter in the universe evolves.

[2]

[B] More massive than Jupiter, brown dwarfs are not massive enough to ⁶⁴fuel the fusion reactions that make stars brightly shine, brown dwarfs are too dim to be observed through telescopes due to their lack of brightness. ⁶⁵

61. A. NO CHANGE
B. when Shiv Kumar, an astronomer who studied
C. Shiv Kumar, an astronomer studying
D. an astronomer, Shiv Kumar, studied
62. F. NO CHANGE
G. dwarfs,” as astronomer, Jill Tarter,
H. dwarfs” as astronomer, Jill Tarter
J. dwarfs” as astronomer Jill Tarter
63. A. NO CHANGE
B. foremost quandaries
C. key questions
D. major basics
64. F. NO CHANGE
G. brown dwarfs do not have enough mass
H. their mass is not enough
J. but not massive enough
65. A. NO CHANGE
B. because they are not as easily seen as stars.
C. despite being quite massive.
D. DELETE the underlined portion (ending the sentence with a period).

Astronomers had to rely on advanced ways of finding
66

these delusive orbs. [C]
67

[3]

Their first clue lay in the process of nuclear fusion.

[D] While brown dwarfs, like stars, initiate fusion, they cannot sustain the constant fusion that causes stars to burn at high temperatures. Still, brown dwarfs lose their heat,
68 cooling over time to temperatures between those of planets and those of stars. An infrared telescope can detect heat signatures in this range, thus helping astronomers home
69 in on potential brown dwarfs.

[4]

With these candidates identified, astronomers were then able to search for a telltale variation in brown dwarfs' chemistry. Fusion reactions rapidly burn through lithium in stars' cores. It follows from the fact that brown dwarfs
70 cannot sustain these reactions, though, most never deplete their lithium supply. An object's spectrum can show which elements, lithium tacked on, remain present. It was by
71 running a spectrum analysis that the first brown dwarfs

were confirmed in 1995—the culmination of three
72 decades of research and collaboration.

66. Which choice best emphasizes that astronomers had no existing methods that would allow them to detect brown dwarfs?

- F. NO CHANGE
- G. devise more sophisticated
- H. use their most creative
- J. pick the newest

67. A. NO CHANGE
B. elucidating
C. elusive
D. allusive

68. F. NO CHANGE
G. In any case,
H. Lastly,
J. Rather,

69. A. NO CHANGE
B. allotment,
C. portion,
D. extent,

70. F. NO CHANGE
G. Logically,
H. Because
J. DELETE the underlined portion (adjusting the capitalization as needed).

71. A. NO CHANGE
B. included,
C. comprised,
D. to boot,

72. If the writer were to delete the underlined portion (ending the sentence with a period), the paragraph would primarily lose a detail that:

- F. emphasizes that locating brown dwarfs relied on years of dedicated work from many contributors.
- G. indicates how long scientists took to confirm that the object discovered in 1995 was a brown dwarf.
- H. conveys the writer's opinion that astronomers should have begun studying brown dwarfs earlier.
- J. suggests that brown dwarfs were not studied until years after they were theorized to exist.



[5]

Since then, hundreds of brown dwarfs have been found and studied, shedding light on how stars, planets, and brown dwarfs differ while also revealing surprising similarities. Some brown dwarfs, for instance, have

planetlike weather patterns—clouds made of
sand—swirl in their atmospheres, bringing rain

73

of liquid iron. 74

73. A. NO CHANGE
B. patterns, clouds made of sand swirl in their atmospheres;
C. patterns: clouds made of sand swirl in their atmospheres,
D. patterns, clouds made of sand, swirl in their atmospheres,
74. Which of the following sentences, if added here, would best conclude the paragraph and the essay?
- F. Young brown dwarfs can actually give off a weak red light that a few astronomers have been lucky enough to spot.
G. Many people wondered if brown dwarfs might even account for the “missing matter” problem in physics.
H. Astronomers hope these are just the first of many discoveries these “invisible” objects will bring to light.
J. Many questions remain, such as how many brown dwarfs the universe holds compared to visible stars.

Question 75 asks about the preceding passage as a whole.

75. The writer wants to add the following sentence to the essay:
- Locating these objects, however, proved difficult.
- This sentence would most logically be placed at:
- A. Point A in Paragraph 1.
B. Point B in Paragraph 2.
C. Point C in Paragraph 2.
D. Point D in Paragraph 3.

END OF TEST 1

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.

**MATHEMATICS TEST***60 Minutes—60 Questions*

DIRECTIONS: Solve each problem, choose the correct answer, and then fill in the corresponding oval on your answer document.

Do not linger over problems that take too much time. Solve as many as you can; then return to the others in the time you have left for this test.

You are permitted to use a calculator on this test. You may use your calculator for any problems you choose,

but some of the problems may best be done without using a calculator.

Note: Unless otherwise stated, all of the following should be assumed.

1. Illustrative figures are NOT necessarily drawn to scale.
2. Geometric figures lie in a plane.
3. The word *line* indicates a straight line.
4. The word *average* indicates arithmetic mean.

1. A point at $(-4,10)$ in the standard (x,y) coordinate plane is translated right 10 coordinate units and down 4 coordinate units. What are the coordinates of the point after the translation?

A. $(-14,14)$
B. $(0, 0)$
C. $(6, 6)$
D. $(6,14)$
E. $(14,14)$

2. A function, f , is defined by $f(x,y) = 3x^2 - 2y$. What is the value of $f(2,3)$?

F. 6
G. 12
H. 23
J. 30
K. 77

3. What is the value of x when $9 - 11x = 13 - 9x$ is true?

A. -2
B. -0.5
C. 0.5
D. 1.1
E. 2

4. A company reimburses its employees for use of a personal vehicle for company travel at the rate of \$10.00 per day plus \$0.445 per mile. Jorge, an employee of the company, traveled exactly 120 miles on Monday and exactly 100 miles on Tuesday using his personal vehicle for company travel. For these 2 days, what was the amount of Jorge's reimbursement?

F. \$ 54.50
G. \$ 63.40
H. \$ 97.90
J. \$107.90
K. \$117.90

DO YOUR FIGURING HERE.



5. Given that $a \triangle b = a^2 + b - \frac{a}{b}$, what is the value of $a \triangle b$ when $a = 5$ and $b = \frac{1}{2}$?

DO YOUR FIGURING HERE.

- A. $\frac{1}{2}$
 - B. 8
 - C. $10\frac{1}{4}$
 - D. $15\frac{1}{2}$
 - E. 23
6. Before opening her lemonade stand, Haley purchased lemonade mix for \$2.30 and a package of cups for \$1.50. Haley sells each cup of lemonade for \$0.50. Which of the following expressions gives Haley's total profit, in dollars, from selling x cups of lemonade?
- F. $0.5x - 3.8$
 - G. $0.5x + 0.8$
 - H. $0.5x + 3.8$
 - J. $0.8x + 0.5$
 - K. $3.8x - 0.5$
7. In $\triangle ABC$, the measure of $\angle B$ is 90° , and the measure of $\angle A$ is 4 times the measure of $\angle C$. What is the measure of $\angle C$?
- A. 18°
 - B. 22.5°
 - C. 36°
 - D. 45°
 - E. 72°
8. The Sheffield High School girls' softball team currently has a record of 11 wins, 8 losses, and 0 ties. What is the *least* number of its remaining 14 games the team must win to finish the season winning *more than* 50% of all the team's games?
- F. 3
 - G. 6
 - H. 7
 - J. 8
 - K. 10
9. A line in the standard (x,y) coordinate plane passes through the point $(3,4)$. The slope of the line:
- A. is positive.
 - B. is zero.
 - C. is negative.
 - D. is undefined.
 - E. cannot be determined from the given information.



10. The probability that a certain basketball player will make any given free throw is 0.8. Based on this probability, how many free throws should this player expect to have to attempt in order to make 20 free throws?

F. 36
G. 28
H. 25
J. 24
K. 22

DO YOUR FIGURING HERE.

11. Jerrica drives to and from school once per day in a car that travels 20 miles on 1 gallon of gas. A trip from Jerrica's house to school and back is 6 miles. Gas costs \$5 per gallon. Suppose Jerrica rides her bike to and from school for 10 school days. How much money will Jerrica save on gas by not driving to and from school those days?

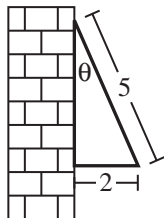
A. \$ 3
B. \$ 6
C. \$15
D. \$24
E. \$30

12. If $x > 0$ and $2x^2 + 5x = 3$, what is the value of x ?

F. 1
G. 3
H. $\frac{1}{2}$
J. $\frac{3}{2}$
K. $\frac{5}{2}$

13. A 5-foot awning extends 2 feet horizontally over the entrance to the vertical building shown below. Which of the following expressions has a value equal to the measure of the angle θ between the building and the awning?

A. $\sin^{-1}\left(\frac{2}{5}\right)$
B. $\cos^{-1}\left(\frac{2}{\sqrt{21}}\right)$
C. $\cos^{-1}\left(\frac{2}{5}\right)$
D. $\tan^{-1}\left(\frac{\sqrt{21}}{2}\right)$
E. $\tan^{-1}\left(\frac{2}{5}\right)$





Use the following information to answer questions 14–16.

DO YOUR FIGURING HERE.

The results for the 200-meter dash at the Shamrock High School (SHS) track meet are shown in the table below. The SHS track is unique in that it is circular and has an outer diameter of 240 meters. The stadium has a capacity of 5,000 people.

200-meter dash	
Name	Time (seconds)
Anthony	21.20
Burke	22.01
Cory	21.33
Kate	22.13
Lindsey	21.20
Nathan	21.50
Rebecca	21.48
Sam	22.19

14. What is the mean, in seconds, of the times listed in the table for the 200-meter dash?
- F. 21.20
G. 21.27
H. 21.49
J. 21.63
K. 22.07
15. Each ticket for the track meet had a price of \$8. Tickets were sold for exactly 85% of the stadium's capacity. How many dollars did SHS collect through ticket sales for this meet?
- A. \$ 3,400
B. \$ 34,000
C. \$ 39,915
D. \$ 40,085
E. \$340,000
16. The circular region containing the track and the infield (the grassy region enclosed by the track) is divided into sectors that each have a central angle of 60° . Each sector is a different color. What is the area, in square meters, of each sector?
- F. $2,400\pi$
G. $4,800\pi$
H. $9,600\pi$
J. $14,400\pi$
K. $19,200\pi$



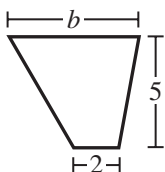
17. The solution set of the system of inequalities below includes which of the following (x,y) pairs?

$$\begin{aligned} x &> y \\ x + y &> 3 \end{aligned}$$

- A. $(0,4)$
- B. $(1,3)$
- C. $(2,1)$
- D. $(3,0)$
- E. $(4,1)$

DO YOUR FIGURING HERE.

18. The area of the trapezoid shown below is 20 square feet, the height is 5 feet, and the length of one base is 2 feet. What is b , the length of the other base, in feet?



- F. 6
- G. 8
- H. 13
- J. 15
- K. 23

19. A square and a rectangle have the same area. The length of the rectangle is 196 centimeters, and the width of the rectangle is 4 centimeters. What is the length, in centimeters, of a side of the square?

- A. 20
- B. 28
- C. 100
- D. 400
- E. 784

20. Let $(3x - 1)(2x - 5) = ax^2 + bx + c$. What is the value of $a + b + c$?

- F. -16
- G. -7
- H. -6
- J. -2
- K. 24

21. For $a \neq 0$ and any real number n , the expression $\frac{a^{2n+1}}{a^n}$ is equivalent to which of the following?

- A. a
- B. a^3
- C. a^n
- D. a^{n+1}
- E. a^{3n+1}



22. For what positive value of a is it true that $(\sqrt{8a})(\sqrt{2a}) = 16$?

F. $\frac{1}{2}$
 G. 1
 H. 2
 J. 4
 K. 16

DO YOUR FIGURING HERE.

23. A basket contains 3 red apples, 4 green apples, and 2 yellow apples. An apple is drawn at random, eaten, and then another apple is randomly drawn. If the first apple is red, what is the probability the second apple is yellow?

A. $\frac{1}{12}$
 B. $\frac{1}{8}$
 C. $\frac{1}{4}$
 D. $\frac{2}{9}$
 E. $\frac{5}{81}$

24. Which of the following numbers is irrational?

F. π
 G. $3\frac{1}{3}$
 H. 2.193876
 J. $\sqrt{144}$
 K. 4

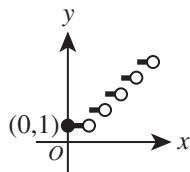
25. The expression $7[3 - 2(x - 4)]$ is equivalent to:

A. $-14x - 35$
 B. $-14x - 7$
 C. $-14x + 77$
 D. $5x - 1$
 E. $5x + 11$

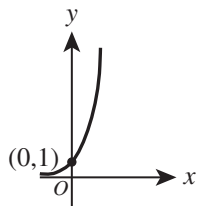


26. Rate of change is defined to be the vertical change between any 2 points divided by the horizontal change between the same 2 points. A function containing the point $(0,1)$ has a constant rate of change equal to $\frac{3}{2}$. The function is shown in one of the standard (x,y) coordinate plane graphs below. Which one?

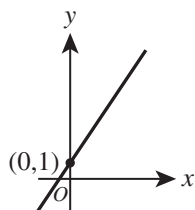
F.



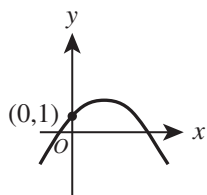
J.



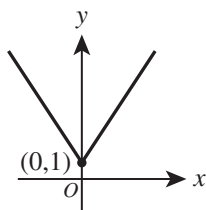
G.



K.



H.



27. What rational number is exactly halfway between $\frac{3}{8}$ and $\frac{7}{16}$ on the real number line?

- A. $\frac{13}{8}$
 B. $\frac{13}{16}$
 C. $\frac{5}{24}$
 D. $\frac{1}{32}$
 E. $\frac{13}{32}$

28. One neon sign flashes every 4 seconds, and another neon sign flashes every 6 seconds. At a certain instant, the 2 signs flash at the same time. How many seconds elapse until the 2 signs next flash at the same time?

- F. 2
 G. 5
 H. 10
 J. 12
 K. 24

DO YOUR FIGURING HERE.

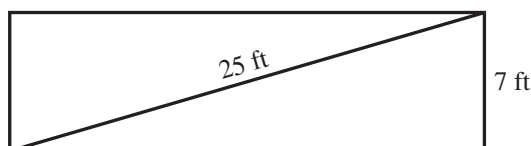


Use the following information to answer questions 29–31.

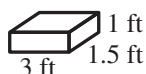
DO YOUR FIGURING HERE.

Jerry is building a rectangular patio against one side of his house. The top surface of the patio, along with its width and diagonal, is shown in the figure below. To build the patio, he will first dig a rectangular hole to a uniform depth of 8 inches. He will then fill the hole with gravel to a uniform depth of 3 inches. He will purchase gravel from his local rock quarry for \$54 per cubic yard (1 cubic yard = 27 cubic feet). Next, he will pour concrete on top of the gravel to obtain a smooth surface for the patio. Finally, he will frame and build a rectangular concrete step with the dimensions shown below.

patio



step



29. Jerry will purchase 42 cubic feet of gravel for his patio. The local rock quarry will sell gravel in the exact volume needed by the customer. What is the cost, in dollars, to purchase the gravel from the local rock quarry?
- A. \$ 54
B. \$ 84
C. \$162
D. \$270
E. \$432
30. What is the area, in square feet, of the top surface of Jerry's patio?
- F. 18
G. 24
H. 168
J. 175
K. 576
31. Jerry will use bags of concrete mix to make the step. Each bag makes 0.4 cubic feet of concrete. To the nearest 0.1 bag, how many bags of concrete mix will he need for the step?
- A. 1.8
B. 4.5
C. 5.5
D. 11.3
E. 18.0



32. A carpenter needs a board $35\frac{5}{8}$ inches long and another board $27\frac{1}{2}$ inches long. Both pieces are to be cut from a single board 72 inches long. Together, the 2 cuts subtract a total of $\frac{1}{8}$ inch. What length of board is left over, in inches?

F. $8\frac{1}{4}$
 G. $8\frac{3}{4}$
 H. $8\frac{13}{16}$
 J. $9\frac{27}{80}$
 K. $9\frac{3}{4}$

33. Given matrices A and B such that $A = \begin{bmatrix} -7 & 2 & 4 \\ -1 & 0 & -3 \end{bmatrix}$ and

$B - A = \begin{bmatrix} 6 & 7 & 4 \\ 1 & -1 & -4 \end{bmatrix}$, what is matrix B ?

A. $\begin{bmatrix} -13 & -5 & 0 \\ -2 & 1 & 1 \end{bmatrix}$
 B. $\begin{bmatrix} -1 & 9 & -8 \\ 0 & 1 & -7 \end{bmatrix}$
 C. $\begin{bmatrix} -1 & 9 & 8 \\ 0 & -1 & -7 \end{bmatrix}$
 D. $\begin{bmatrix} 13 & 5 & 0 \\ 2 & -1 & -7 \end{bmatrix}$
 E. $\begin{bmatrix} 13 & 9 & 8 \\ 2 & 1 & 7 \end{bmatrix}$

34. A Chinese language proficiency test was given to 60 university students. The students' scores were recorded as integers. The table below shows the frequency and cumulative frequency of test scores in certain ranges.

Range of scores	Frequency	Cumulative frequency
50–59	6	6
60–69	16	22
70–79	16	38
80–89	12	50
90–100	10	60

To be classified as *proficient*, a student must score 70 or higher on the test. What is the ratio of the number of students who were classified as proficient to the number who were NOT classified as proficient?

F. 19:3
 G. 19:11
 H. 25:11
 J. 30:11
 K. 30:19

DO YOUR FIGURING HERE.



35. A group of 4 friends decide to go to a movie. After buying their tickets, they find 5 empty chairs in the theater. How many different seating arrangements are possible if the 4 friends sit, at most 1 person per chair, in these 5 chairs?

A. 5
 B. 15
 C. 24
 D. 120
 E. 240

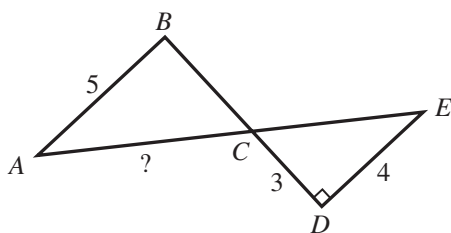
DO YOUR FIGURING HERE.

36. Carina rolls 2 cubes. Each has the whole numbers from 1 through 6 on its faces, 1 per face. The faces of each cube are equally likely to land up. What is the probability that the sum of the numbers on the faces landing up is 3 or more?

F. $\frac{1}{36}$
 G. $\frac{1}{18}$
 H. $\frac{1}{11}$
 J. $\frac{10}{11}$
 K. $\frac{35}{36}$

37. In the figure shown below, $\overline{AB} \parallel \overline{DE}$, C is the intersection of $\overline{AE}$ and $\overline{BD}$, $\angle D$ is a right angle, and the given side lengths are in inches. What is the length of $\overline{AC}$, in inches?

A. $\frac{15}{4}$
 B. 4
 C. 5
 D. $\frac{25}{4}$
 E. $\frac{25}{3}$



38. The domain of the real-valued function $f(x) = \frac{2}{\sqrt{x-3}}$ is the set of all x -values that satisfy:

F. $x > 0$
 G. $x \geq 0$
 H. $x > 2$
 J. $x > 3$
 K. $x \neq 3$



39. For $f(x) = 2x + 3$ and $g(x) = x^2 - 4$, which of the following expressions represents $f(g(x))$?

A. $2x^2 - 1$
 B. $2x^2 - 5$
 C. $2x^2 - 8$
 D. $4x^2 + 5$
 E. $4x^2 + 12x + 5$

DO YOUR FIGURING HERE.

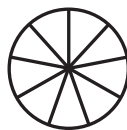
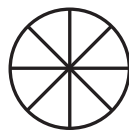
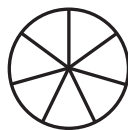
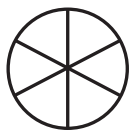
40. For all $y > 0$, which of the following is a simplified form of $\frac{y-1}{y+1} - \frac{2(y-1)}{4y+4}$?

F. $-\frac{1}{2}$
 G. 0
 H. $\frac{1}{2}$
 J. $\frac{y-1}{3(y+1)}$
 K. $\frac{y-1}{2(y+1)}$

41. A wheel with a radius of 1 centimeter rolls without slipping along the ground. After exactly 2 revolutions, how many centimeters has the wheel traveled along the ground?

A. 2
 B. π
 C. 2π
 D. 4
 E. 4π

42. For Cameron's birthday party, his mom bought 4 pizzas, each the same size but with a different topping. She cut those pizzas into 6, 7, 8, and 9 equal pieces, respectively, as shown below, giving 5 slices to each of the 6 boys attending. If each boy ate all 5 of his slices and had at least 3 different kinds of pizza, which of the following expressions gives the largest amount of pizza that 1 boy could have eaten (expressed in fractions of a pizza)?



F. $\frac{1}{7} + \frac{1}{8} + \frac{3}{9}$
 G. $\frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \frac{2}{9}$
 H. $\frac{2}{6} + \frac{1}{7} + \frac{1}{8} + \frac{1}{9}$
 J. $\frac{2}{6} + \frac{2}{7} + \frac{1}{8}$
 K. $\frac{3}{6} + \frac{1}{7} + \frac{1}{8}$

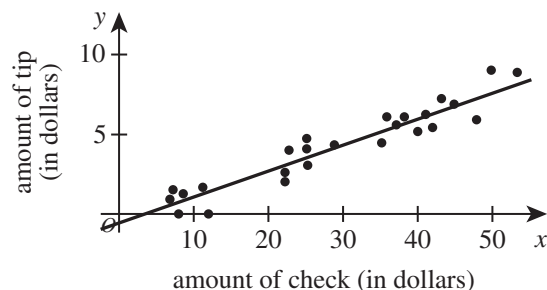


43. When x , y , and z are positive integers, which of the following relationships will assure that the product $x^0y^1z^{-1}$ will have a value greater than 1 ?

A. $x < y$
 B. $x > y$
 C. $x > z$
 D. $y < z$
 E. $y > z$

DO YOUR FIGURING HERE.

44. Sumiko, who works at a restaurant, recorded the amounts of her customers' checks and the corresponding amounts of her tips, both in dollars, for 25 consecutive customers she served. She plotted the data in the standard (x,y) coordinate plane below with the amount of the check on the x -axis and the amount of the tip on the y -axis. Sumiko then performed a linear regression on the data, finding the regression equation to be $y = 0.16x - 0.44$ with a correlation coefficient, r , of 0.81. She also graphed the regression line.



From her statistical analysis, Sumiko can correctly conclude that every \$1 increase in the total amount of the check will result in approximately:

- F. a 16 cent increase in her tip.
 G. a 44 cent increase in her tip.
 H. an 81 cent increase in her tip.
 J. a 16 cent decrease in her tip.
 K. a 44 cent decrease in her tip.
45. The cost of a certain type of carpet is \$10 per square yard. What would be the cost of carpet of this type to cover a rectangular floor 12 feet by 18 feet ?
- A. \$ 216
 B. \$ 240
 C. \$ 600
 D. \$ 720
 E. \$2,160



46. The function $f(x) = x^5$ is defined for all real numbers x . Which of the following expressions represents $f^{-1}(x)$?

F. x^{-5}

G. $x^{-\frac{1}{5}}$

H. $x^{\frac{1}{5}}$

J. $\frac{x}{5}$

K. $-\frac{x}{5}$

DO YOUR FIGURING HERE.

47. A car accelerates at a constant rate from 0 miles per hour to 60 miles per hour in 15 seconds and averages 30 miles per hour over the 15-second period. The car maintains the constant speed of 60 miles per hour for the next 15 seconds. What fraction of a mile has the car traveled in these 30 seconds?

A. $\frac{1}{8}$

B. $\frac{1}{4}$

C. $\frac{3}{8}$

D. $\frac{1}{2}$

E. $\frac{3}{4}$

48. If $a - b = 5$ and $c = 7a - 9 - 7b$, then $c = ?$

F. 44

G. 26

H. -2

J. -4

K. Cannot be determined from the given information

49. If $\sin \theta = \frac{1}{2}$ and $\tan \theta = -\frac{\sqrt{3}}{3}$, then $\cos \theta = ?$

A. $-\frac{2\sqrt{3}}{3}$

B. $-\frac{\sqrt{3}}{2}$

C. $-\frac{\sqrt{3}}{6}$

D. $\frac{1}{2}$

E. $\frac{\sqrt{3}}{2}$



50. The graph in the standard (x,y) coordinate plane of $y = a(x - b)^2$, where $|a| \neq |b|$, is reflected across the x -axis. Which of the following is an equation of the reflection?

F. $y = -a(x - b)^2$
 G. $y = a(x - b)^2$
 H. $y = a(x + b)^2$
 J. $x = -b(y - a)^2$
 K. $x = b(y + a)^2$

DO YOUR FIGURING HERE.

51. For $i^2 = -1$, $\frac{2+3i}{i} = ?$

A. 5
 B. $5i$
 C. $-3 + 2i$
 D. $3 - 2i$
 E. $3 + 2i$

52. All 30 of Mr. Gray's algebra students took the midterm test, and Mr. Gray computed the average test score for the class. Then, one of Mr. Gray's students, Mandy, found that her test should have been scored 5 points higher than it was originally scored. All other students' scores were correct. After correcting the scoring error on Mandy's test, the average test score for the class increased by how many points?

F. $\frac{1}{30}$
 G. $\frac{1}{6}$
 H. $\frac{1}{5}$
 J. 5

K. Cannot be determined from the given information

53. What is the value of z in the solution to the system of equations below?

$$\begin{aligned} 2x + y - z &= 0 \\ x - 3y + z &= 18 \\ x + y &= 1 \end{aligned}$$

A. -3
 B. $1\frac{1}{2}$
 C. 4
 D. 5
 E. 19

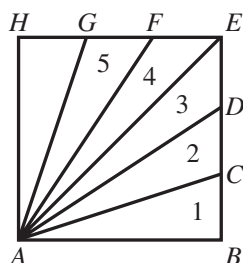


54. What is the maximum possible value of $\frac{x}{y^2}$ for x in the set $\{1, \frac{1}{2}, 0\}$ and y in the set $\{\frac{1}{8}, \frac{1}{4}, \frac{1}{2}\}$?

F. 2
G. 4
H. 8
J. 32
K. 64

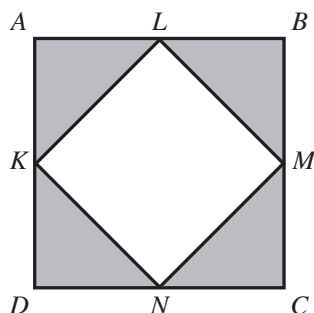
DO YOUR FIGURING HERE.

55. In the figure below, $ABEH$ is a square. Points C and D are on $\overline{BE}$ such that $\overline{BC}$, $\overline{CD}$, and $\overline{DE}$ all have the same length. Points F and G are on $\overline{EH}$ such that $\overline{EF}$, $\overline{FG}$, and $\overline{GH}$ all have the same length. How many of the triangles numbered 1–5 ($\triangle ABC$, $\triangle ACD$, $\triangle ADE$, $\triangle AEF$, and $\triangle AFG$, respectively) have the same area as $\triangle AGH$?



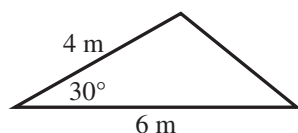
A. 1
B. 2
C. 3
D. 4
E. 5

56. The midpoints of the sides of square $ABCD$ are the vertices of square $KLMN$, shown below. The side length of square $ABCD$ is 14 meters. What is the area, in square meters, of the shaded portion?



F. $7\sqrt{2}$
G. $14\sqrt{2}$
H. 24.5
J. 49
K. 98

57. Given $\sin 30^\circ = \frac{1}{2}$, what is the area, in square meters, of the triangle shown below?



A. 6
B. 8
C. 10
D. 12
E. 24



58. The graph of which of the equations below is a hyperbola with the same asymptotes as the graph of $\frac{x^2}{9} - \frac{y^2}{4} = 1$?

F. $\frac{x^2}{9} + \frac{y^2}{4} = 1$

G. $\frac{x^2}{4} - \frac{y^2}{9} = 1$

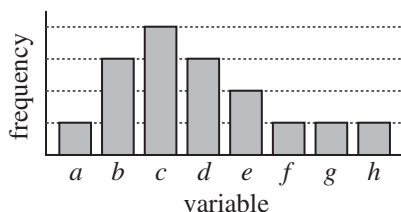
H. $\frac{y^2}{9} + \frac{x^2}{4} = 1$

J. $\frac{y^2}{9} - \frac{x^2}{4} = 1$

K. $\frac{y^2}{4} - \frac{x^2}{9} = 1$

DO YOUR FIGURING HERE.

59. The graph below shows the distribution of a data set. The variables a – h represent 8 consecutive positive integers in order from least to greatest. The frequencies of a , f , g , and h are equal; the frequency of e is 2 times the frequency of a ; the frequency of b and of d is 3 times the frequency of a ; and the frequency of c is 4 times the frequency of a . Which of the following statements about the mean, median, and mode of the data set is true?



- A. The mode is less than the median, and the median is less than the mean.
- B. The mean is less than the median, and the median is less than the mode.
- C. The mode is equal to the mean, and the mean is less than the median.
- D. The mode is less than the mean, and the mean is equal to the median.
- E. The mean is equal to the median, and the median is equal to the mode.
60. A solid rectangular block of steel weighs w pounds. Which of the following expressions gives the weight, in pounds, of a larger solid rectangular block that is made of the same kind of steel and has length, width, and height 4 times those of the smaller block?
(Note: Assume that weight is proportional to volume.)
- F. $4w^3$
- G. $4w$
- H. $16w$
- J. $64w$
- K. $64w^3$

END OF TEST 2

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.

DO NOT RETURN TO THE PREVIOUS TEST.

READING TEST

35 Minutes—40 Questions

DIRECTIONS: There are several passages in this test. Each passage is accompanied by several questions. After reading a passage, choose the best answer to each question and fill in the corresponding oval on your answer document. You may refer to the passages as often as necessary.

Passage I

LITERARY NARRATIVE: This passage is adapted from the autobiographical essay “Song for My Father” by Eric Liu (©1998 by Eric Liu).

It is unlike any other book I own. On the cover is a color photograph of my father. In the photograph, taken sometime in the 1960s, my father’s head is turned to his left, his mouth slightly open in a relaxed smile. Even
5 behind heavy-framed glasses, his eyes appear to be seeing something clearly. It seems he might be saying something soon, something thoughtful, or maybe playful. His skin, still smooth and full, tells me he was a young man not that many years before the picture was
10 taken. But his visage—knowing, kind, self-aware—tells me he has already become the man I knew as Baba. That picture is why I keep the paperback at my bedside. It keeps my father close, sets his gaze upon me as I sleep.

15 The book was compiled by several of my father’s childhood friends after he died in 1991. This wasn’t, as far as I know, some sort of Chinese tradition, publishing a memorial book for a departed chum. It was just an act of loyalty; of love, if I may say that. In part, the
20 book is a record of grief, containing the obituary from the *Poughkeepsie Journal*, my eulogy, an elegiac essay by my mother. But for most of its 198 pages, it is actually a prose reunion, a memoir of the idyllic adolescence of a band of boys in post–World War II
25 Taiwan.

There are pieces in the book, written by my father and his brothers and his classmates, about high school life, about a favorite teacher, about camping and fishing trips, about picaresque adventures where nary an adult
30 appears. There are photographs too; in many of them, Dad and his friends are wearing their school uniforms, baggy and vaguely military. One snapshot I remember vividly. Eight or nine of them are walking up a dirt road, jesting and smiling. And there’s my father at the
35 end of this happy phalanx—khaki hat a bit too big, arm pumping jauntily and foot raised in mid-march, singing a song. The face is my father’s, but the stance, so utterly carefree, is hardly recognizable. I stared at that picture for a long time when I first got the book.

40 It’s through these photographs that I’ll read the book every so often, searching the scenes for new revelations. That’s partly because the photographs are so

wonderful. But it’s also, frankly, because I do not understand the text. Almost all the entries, you see, are
45 written in Chinese—a language that I once could read and write with middling proficiency but have since let slip into disuse. Though I know enough to read from top to bottom, right to left, and “back” to “front,” I recognize so few of the characters now that perusing the
50 text yields little more than frustration, and shame. I know what the book contains only because Mom has told me. And she’s had to tell me several times.

On one or two occasions I’ve sat down with my pocket Chinese-English dictionary, determined to decipher at least the essays that my father wrote. This was
55 painstaking work and I never got very far. For each Chinese character, I first had to determine the ideographic root, then count the brush strokes, then turn to an index ordered by root and by number of strokes, then match the character, then figure out its romanized
60 spelling, then look up its definition. By the time I solved one word, I’d already forgotten the previous one. Meaning was hard enough to determine; context was even more elusive.

65 So it is, I sometimes think, with my father’s life. On the one hand, it’s easy to locate my father and my family in the grand narrative of “the Chinese American experience.” On the other hand, it doesn’t take long for this narrative to seem more like a riddle than a fable.
70 Leafing through the pages of the memorial book, staring dumbly at their blur of ideographs, I realize just how little I know about those years of Baba’s life before he arrived in America, and before I arrived in the world. I sense how difficult it is to be literate in
75 another man’s life, how opaque an inheritance one’s identity truly is. I begin to perceive my own ignorance of self.

When Chao-hua Liu came to the United States in 1955, at the age of eighteen, he was Chinese. When he
80 died thirty-six years later, he was, I’d say, something other than Chinese. And he had helped raise a son who was Chinese in perhaps only a nominal sense. But what, ultimately, does all this mean? Where does this Chineseness reside? In the word? In the deed? In what is
85 learned—or what is already known? And how is it passed from one generation to the next? Some of the answers lie, I know, in a book I am still unable to read. But there are other answers, I suspect, in a book I must now begin to write.

1. Which of the following statements best captures a main theme of the passage?
 - A. It's difficult to know the complexities of another person's life.
 - B. It's easier to deeply know family than it is to deeply know friends.
 - C. Photographs give a clear and vivid view of the subtleties of an ancestor's life.
 - D. Family history seems like a riddle when good records are not kept.
2. Which of the following quotations from the passage contains details that clearly convey what the narrator refers to as his father's "idyllic adolescence" (lines 23–24)?
 - F. "It seems he might be saying something soon, something thoughtful, or maybe playful" (lines 6–8).
 - G. "His visage . . . tells me he has already become the man I knew as Baba" (lines 10–11).
 - H. "There are pieces . . . about picaresque adventures where nary an adult appears" (lines 26–30).
 - J. "It doesn't take long for this narrative to seem more like a riddle than a fable" (lines 68–69).
3. Details in the first paragraph make clear that the narrator views the color photograph of his father with:
 - A. fondness and respect.
 - B. wonder and skepticism.
 - C. curiosity and despair.
 - D. apprehension and sadness.
4. It can most reasonably be inferred that the narrator stared for a long time at the snapshot of his father that's discussed in the third paragraph (lines 26–39) because:
 - F. the photograph has aged so badly that it's unclear who the people in the photo are.
 - G. the photograph reveals an aspect of his father's personality that the narrator has never before seen.
 - H. his father's face looks young in the photograph given that the photo was taken with an old-fashioned camera.
 - J. his father would later barely resemble the photograph in any way.
5. The main idea of the fifth paragraph (lines 53–64) is that the narrator's attempt to read the text in the memorial book was:
 - A. difficult but exhilarating.
 - B. exciting and productive.
 - C. exhausting but successful.
 - D. time consuming and fruitless.
6. In the passage, as the narrator looks through the memorial book and considers his father's life before his father came to the United States, the narrator becomes increasingly:
 - F. disappointed that his father had rarely talked about his experiences during World War II.
 - G. aware of his own ignorance of self and how opaque an identity truly is.
 - H. proud of his father's decision to leave Taiwan and live in America.
 - J. closer to understanding the riddle that was his father's life.
7. The last paragraph primarily serves to:
 - A. raise unresolved issues that the narrator hopes to find answers to.
 - B. list the questions that the memorial book, if translated, could answer.
 - C. argue the narrator's view that he is Chinese in more than just a nominal sense.
 - D. outline the narrator's father's life, as it is portrayed in the memorial book.
8. In terms of learning the Chinese language, the narrator characterizes himself as someone who has:
 - F. never been able to read and write Chinese, even though it's the language of his parents.
 - G. often forgotten to read from right to left and "back" to "front" while reading Chinese.
 - H. gained considerable skill in reading Chinese through studying the memorial book.
 - J. all but lost the middling ability he once had to read and write Chinese.
9. In regard to the memorial book, the narrator indicates that his mother has:
 - A. struggled to remember what she has already told him about it.
 - B. shared its contents with him on several occasions.
 - C. pushed him to learn how to read it on his own.
 - D. dismissed his interest in understanding its contents.
10. The narrator states that in terms of Chao-hua Liu's cultural identity, late in his life Chao-hua Liu had become:
 - F. primarily Chinese.
 - G. primarily American.
 - H. something other than Chinese.
 - J. more Chinese than when he came to the United States.

Passage II

SOCIAL SCIENCE: Passage A is adapted from the article “At Libraries, Quiet Makes a Comeback” by Jennifer Howard (©2012 by The Chronicle of Higher Education, Inc.). Passage B is adapted from the book *The Pleasures of Reading in an Age of Distraction* by Alan Jacobs (©2011 by Alan Jacobs).

Passage A by Jennifer Howard

The buzzing of smartphones, the clacking of computer keys, the chatter of study groups: Academic libraries aren’t the quiet temples to scholarship they used to be. Personal portable technology takes some of the blame. So does the current pedagogical emphasis on group work. In response to students’ devices and habits, many college libraries have installed coffee shops and embraced the learning-commons model of design, creating wired spaces where groups can gather and plug in.

Library quiet is making a comeback, though, in part because students themselves are asking for it. “Students crave quiet as much as they crave conversation,” says Karen G. Schneider, director for library services at Holy Names University. “A lot of libraries are zoning their space into areas where there can be quiet conversation, absolute silence, and even livelier conversations.”

The way librarians, students, and researchers behave undermines what Ms. Schneider calls “the stereotype of the shushing librarian and absolute pin-drop silence.” She and other librarians point out that students are often the first to bring up noise problems in the library, to ask for more quiet spaces, and to police those spaces themselves.

“We assume this is the multitasking generation. It’s how they’ve been labeled,” says Alison J. Head, co-director of the University of Washington’s Project Information Literacy, which looks at how young adults do research. But her own research has shown that “these students do seek solitude, they do seek the quaint hush of the library and a place they can dial down.”

Eric A. Kidwell, director of the library at Huntingdon College, cites “this prevailing attitude among faculty and administrators that all students want to be in a group situation all the time.” That’s not the case, in his experience. “Maybe there’s more of that, but there are plenty of students out there who enjoy being by themselves, or can’t study with their earphones in and the TV on.”

The solution, in Mr. Kidwell’s view, is “to be as creative as we can in our space design to provide as many different types of reading and study space as we can.” The most popular room at his library is not a fully wired collaborative space but a small room that’s designed to look like an old-fashioned study. “We bought a small table, a rocking chair, a wingback chair,” he says. “We wish we had the ability to make more rooms like that.”

Passage B by Alan Jacobs

The noisier the environment, the more readers are driven to be silent. The boy trying to study at the kitchen table while the clamor of family life goes on around him is a typical figure in the history of reading. No one could plausibly claim that we late moderns are uniquely challenged in this respect: surely a higher percentage of human beings today have regular access to silence than at any time in human history. Most Americans and Western Europeans, and many people elsewhere—not all, mind you—live in environments with quiet rooms, or quiet corners. And many who lack quiet homes have had access to libraries, which have for centuries been dedicated, as it were, to silence.

That may be changing, though. Here’s an example from the United Kingdom that has analogues in several other countries. In 2008, when Andy Burnham was Gordon Brown’s Secretary of State for Culture, Media and Sport, he announced that libraries had to change: “Libraries should be a place for families and joy and chatter. The word chatter might strike fear into the heart of traditionalists but libraries should be social places that offer an antidote to the isolation of someone playing on the internet at home.” Libraries need to “look beyond the bookcase,” which means, among other things, allowing the use of cellphones and selling (or at least allowing people to bring in) food and drink.

Even among educated folk, these proposals got some support, largely out of fear that without some such change libraries will become irrelevant and unused, and eventually will disappear altogether. But others fumed. Victoria Coren wrote, “My brain is struggling with the philosophical question: if there is nothing inside but people eating burgers and playing computer games, is it actually a library? Isn’t it just an internet cafe?” To the “they must look beyond the bookcase” line, Coren retorted, “Why? Why must they? They’re libraries. The bookcase is exactly where they should be looking. Someone should be. I can eat and make phone calls at home.”

Questions 11–13 ask about Passage A.

11. Kidwell’s solution, as it is presented in the sixth paragraph of Passage A (lines 40–48), is intended to address which of the following problems?

- A. Library materials need to be useful for students doing a variety of research.
- B. College students are using study spaces in libraries more than they used to.
- C. Libraries often lack the space to provide new materials.
- D. College students who use libraries have varying study preferences and needs.

12. Passage A states that college libraries have responded to changes in technology and students' habits by:
- F. adding more desks for computers and extending library hours.
 - G. installing coffee shops and creating wired spaces for group work.
 - H. offering wireless Internet access and teaching classes on effective research.
 - J. creating wired study rooms and setting up technology support centers.

13. In the context of Passage A, the idea that "all students want to be in a group situation all the time" (lines 34–35) most closely reflects the perspective of:
- A. library directors.
 - B. psychology scholars.
 - C. faculty and administrators.
 - D. college students and professors.

Questions 14–17 ask about Passage B.

14. Based on Passage B, which of the following activities, were it to occur in a library, would most directly conflict with Burnham's ideas about libraries?
- F. Using cell phones near quiet areas
 - G. Playing games with friends
 - H. Browsing the Internet alone
 - J. Discussing a book with coworkers
15. In Passage B, the example of a boy trying to study at the kitchen table most nearly serves to make the point that:
- A. some people don't need a quiet environment in order to read effectively.
 - B. families can help children read by reducing noise in the home.
 - C. children used to have better access to quiet places than they do today.
 - D. readers have always contended with noisy environments.
16. As it is used in line 54, the word *respect* most nearly means:
- F. regard.
 - G. esteem.
 - H. attention.
 - J. honor.

17. According to Passage B, "educated folk" (line 75) showed support for Burnham's proposals mainly because these people:

- A. thought the proposals would allow libraries to expand their book collections.
- B. wanted to ensure that all people had a place where they could relax and converse.
- C. believed the proposals might result in government programs to fund libraries.
- D. worried that libraries might otherwise become unused and eventually disappear.

Questions 18–20 ask about both passages.

18. Would Coren, as she is presented in Passage B, most likely approve of using the learning-commons model of design in libraries as it is described in Passage A?
- F. Yes, because the learning-commons model focuses on keeping books in libraries.
 - G. Yes, because the learning-commons model allows people to eat and drink in the library.
 - H. No, because the learning-commons model establishes silent zones in libraries.
 - J. No, because the learning-commons model promotes technology and change in libraries.
19. Based on Head's research as it is presented in Passage A, how would college students most likely react if libraries were to adopt Burnham's ideas as they are presented in Passage B?
- A. Students would come to libraries to escape the solitude of home.
 - B. Students would congregate in areas of the library where conversation is encouraged.
 - C. Students would try to find areas of the library that still offer quiet and solitude.
 - D. Students would welcome the chance to play Internet games in the quiet of the library.
20. Which of the following quotations from Passage B most nearly echoes Kidwell's comment about college students in lines 36–39 of Passage A?
- F. "The noisier the environment, the more readers are driven to be silent" (lines 49–50).
 - G. "Libraries should be a place for families and joy and chatter" (lines 67–68).
 - H. "Without some such change libraries will become irrelevant and unused" (lines 76–78).
 - J. "The bookcase is exactly where they should be looking" (lines 85–86).

Passage III

HUMANITIES: This passage is adapted from the book *Becoming Mona Lisa: The Making of a Global Icon* by Donald Sassoon (©2001 by Donald Sassoon).

The popularity of the *Mona Lisa* is not just modern hype. This is an innovative and revolutionary painting. Taken one by one, few of the innovations deployed in the *Mona Lisa* were originally devised by Leonardo da Vinci. As in other fields (literature, philosophy, cinema) there is a difference between being the first to produce a novelty and being the first to understand fully its implications and be able to use them in a masterly fashion. The pose of the *Mona Lisa* was one such significant innovation. She sits presenting a three-quarter view, while the face looks in a different direction—the so-called *contrapposto* position. She looks directly at the viewer, another innovation. Leonardo had broken his own rules, for he had written that women should be painted in a demure position, ‘their heads lowered and inclined to one side.’ Titian and other Italian painters often painted women with downcast eyes, as if they avoided the insolence of the male gaze. *Mona Lisa* looks back, defiantly.

The translucent veils add depth to the painting. The numerous ripples of the garments, the rivulets of drapery falling from her shoulders and the spiral folds of the mantle across the left breast highlight the close spatial relationship between the sitter and the landscape in a way that was very unusual. The barely perceptible columns at either side frame the portrait. The background is asymmetrical, the right side being a little higher than the left.

In traditional Renaissance portraiture, the main function of a landscape was to contribute additional elements for the identification of the sitter—for instance, the Tuscan countryside, or an urban setting. The landscape background of the *Mona Lisa* is not easily identifiable. Behind the balcony on which the woman is sitting are rocks, lakes, streams, but no trees or plants. The sole human structure is a bridge on the right-hand side. In portraits, such completely imaginary backgrounds are rare. The result of the complex system of asymmetric relations is to produce the idea of movement.

The *Mona Lisa* constitutes a major development in the use of the *contrapposto* technique in paintings. The *contrapposto* was an attempt to solve a riddle that had baffled painters for a long time: how to represent on two dimensions (the flat surface) the three-dimensional conception of movement. By definition, a pose is static: the sitter remains still in order to enable the painter to draw. To represent the idea of movement, the static pose of the sitter must constitute a single moment in a sequence—like a photograph of an object in motion. By having the visage of the sitter facing a point different from that facing the torso, the entire body appears to be in motion. This structure had been used often in complex works with several figures, but seldom in one with

a single sitter, as is the case with *Mona Lisa*. She is now a person in motion. In our age, the *contrapposto* position is regularly used by ordinary photographers, who tell their sitters to look away from the direction in which the body is facing so as to avoid the rigidity and stillness of body and face looking symmetrically towards the camera.

The *Mona Lisa* was Leonardo’s solution to a series of technical experimentation he had been working on for decades, the culmination of a lengthy artistic itinerary.

The technique used to give the impression of movement was impressive. But Leonardo used other innovations. He was an early convert to the use of oil paint in place of the quick-drying tempera—the technique of using egg yolk to bind the paint pigments together. Oil paint, slower to dry, appealed to perfectionists. It enhanced the luminosity of the painting and permitted a more detailed and accurate representation. It was first used in northern Europe, and reached Venice and then the rest of Italy in the first half of the fifteenth century.

The northern European connection, though very important, was not one-way. Albrecht Dürer sojourned in Venice repeatedly to learn the techniques of the Italians, who in turn were dazzled by the inventiveness of his prints. Antonello da Messina was supposed to have studied in Bruges with the Van Eyck brothers and brought back to Venice the innovative use of oil paint. Indeed Vasari attributes to ‘Giovanni da Bruggia’ (John of Bruges, i.e. Jan Van Eyck) the invention of oil paint. This is unlikely, but the story illustrates the considerable cultural interchange in fifteenth-century Europe. Antonello had preceded Leonardo in Milan by some ten years and had used both the *contrapposto* and the direct look.

21. Based on the passage, the choice Leonardo made regarding how to depict *Mona Lisa*’s gaze could be considered:

- A. symbolic; *Mona Lisa* stands for the typical Renaissance woman.
- B. unexpected; it broke Leonardo’s own rule that women should pose demurely.
- C. unsurprising; it was the choice Italian painters traditionally had made.
- D. foolish; it caused Leonardo to be mocked by his contemporaries.

22. As it is used in line 3, the word *deployed* most nearly means:
- F. released.
 - G. utilized.
 - H. diverged.
 - J. ordered.
23. According to the passage, in traditional Renaissance portraiture, what role did the landscape behind the sitter generally play?
- A. It included elements such as rocks, lakes, and streams to create a pastoral setting.
 - B. It created a stable, symmetrical background for a sitter who was posed asymmetrically.
 - C. It highlighted a close spatial relationship between the sitter and the background.
 - D. It incorporated details from the sitter's life to further identify whom the portrait depicted.
24. According to the passage, the background of the *Mona Lisa* differs from traditional fifteenth-century portrait backgrounds in that it depicts:
- F. an imaginary setting.
 - G. a structure built by humans.
 - H. the Tuscan countryside.
 - J. both rural and urban elements.
25. According to the passage, the technique Leonardo perfected in the *Mona Lisa* that makes a subject appear to be in motion had often been used before in paintings that:
- A. featured several figures.
 - B. featured landscape backgrounds.
 - C. were created by the Van Eyck brothers.
 - D. were designed by artists using tempera paint.
26. The passage directly refers to the use of the *contrapposto* position today by:
- F. dancers.
 - G. photographers.
 - H. filmmakers.
 - J. cartoonists.
27. Based on the passage, which of the following descriptions best captures what it means for a figure in a painting to be in the *contrapposto* position?
- A. The body and face of the figure point in different directions.
 - B. The figure appears serene against an unsettling or mysterious background.
 - C. The gaze of the figure is downcast as if to keep the viewer at a distance.
 - D. The figure is more prominent than other elements, such as landscape.
28. In the passage, Leonardo's "lengthy artistic itinerary" (lines 64–65) is a reference to:
- F. the trip Leonardo took throughout northern Europe to study oil paint.
 - G. Leonardo's sojourn in Venice with Dürer to learn Italian artistic techniques.
 - H. a list of artistic riddles that Leonardo was famous for having solved.
 - J. the figurative journey Leonardo took as he experimented with artistic techniques.
29. According to the passage, during the time Leonardo was creating works of art, egg yolks were used in paint to:
- A. highlight the smallest details in the painting.
 - B. enhance the luminosity of the painting.
 - C. keep the paint from drying too quickly.
 - D. bind the paint pigments together.
30. It can most reasonably be inferred from the passage that oil paint appealed to artists who were perfectionists because oil paint:
- F. dried slowly, leading to a less luminous surface and therefore images that were more realistic.
 - G. enabled them to produce more accurate representations through the use of fine detail.
 - H. was rare, giving painters who used it license to experiment without inviting criticism.
 - J. had the reputation of being used by only the finest painters in northern Europe.

Passage IV

NATURAL SCIENCE: This passage is adapted from the article “Dinosaurs’ Living Descendants” by Richard Stone (©2010 by Smithsonian Institution).

A Chinese farmer, Li Yingfang, made one of the greatest fossil finds of all time, in August 1996 in Sihetun village. “I was digging holes for planting trees,” recalls Li. From a hole he unearthed a two-foot-
5 long shale slab. An experienced fossil hunter, Li split the slab and beheld a creature unlike any he had seen. The skeleton had a bird-like skull, a long tail and impressions of what appeared to be feather-like structures.

10 Because of the feathers, Ji Qiang, then the director of the National Geological Museum, which bought one of Li’s slabs, assumed it was a new species of primitive bird. But other Chinese paleontologists were convinced it was a dinosaur.

15 On a visit to Beijing that October, Philip Currie, a paleontologist now at the University of Alberta, saw the specimen and realized it would turn paleontology on its head. The next month, Currie showed a photograph of it to colleagues at the annual meeting of the Society of
20 Vertebrate Paleontology. The picture stole the show.

Despite the feathers, the skeleton left no doubt that the new species, named *Sinosauropteryx*, meaning “Chinese lizard wing,” was a dinosaur. Its integumentary filaments—long, thin structures protruding from its
25 scaly skin—convinced most paleontologists that the animal was the first feathered dinosaur ever unearthed. A dozen dinosaurs with filaments or feathers have since been discovered at that site.

The abundance of feathered fossils has allowed
30 paleontologists to examine a fundamental question: Why did feathers evolve? Today, it’s clear that feathers perform many functions: they help birds retain body heat, repel water and attract a mate. And of course they aid flight—but not always, as ostriches and penguins,
35 which have feathers but do not fly, demonstrate. Many feathered dinosaurs did not have wings or were too heavy, relative to the length of their feathered limbs, to fly.

Deciphering how feathers morphed over the ages
40 from spindly fibers to delicate instruments of flight would shed light on the transition of dinosaurs to birds, and how natural selection forged this complex trait. Few scientists know ancient feathers more intimately than Xu Xing of the Institute of Vertebrate Paleontology and Paleoanthropology (IVPP) in Beijing. He has
45 discovered 40 dinosaur species—more than any other living scientist—from all over China.

Xu envisions feather evolution as an incremental process. Feathers in their most primitive form were
50 single filaments, resembling quills, that jutted from reptilian skin. These simple structures go way back; even pterodactyls had filaments of sorts.

Originally, single filaments may well have been for display, the dinosaur equivalent of a peacock’s
55 iridescent plumage. Vivid evidence for that theory appeared when scientists unveiled the true colors of 125-million-year-old feathers. Bird feathers and reptile scales contain melanosomes—tiny sacs holding varieties of the pigment melanin. Many paleontologists
60 suspected that dinosaur feathers also contained melanosomes. In Mike Benton’s laboratory at the University of Bristol, IVPP’s Zhang Fucheng spent more than a year searching for melanosomes in photographs of bird and dinosaur fossils taken with an electron
65 microscope. Zhang’s diligence paid off in 2009 when he pinpointed melanosomes in the primitive bird species *Confuciusornis* that contained eumelanin, which gives feathers a gray or black tinge, and pheo-
70 melanin, which gives them a chestnut to reddish-brown color. The animal’s feathers had patches of white, black and orange-brown coloring.

Sinosauropteryx was even more stunning. Zhang found that the filaments running down its back and tail must have made the dinosaur look like an orange-and-
75 white-striped barber pole. Such a vibrant pattern suggests that “feathers first arose as agents for color display,” Benton says.

Feathers did, of course, eventually become an instrument of flight. Some paleontologists envision a
80 scenario in which dinosaurs used feathers to help them occupy trees for the first time. “Because dinosaurs had hinged ankles, they could not rotate their feet and they couldn’t climb well. Maybe feathers helped them scramble up tree trunks,” says paleontologist Michael
85 Carrano. Baby birds of primarily ground-dwelling species like turkeys use their wings in this way. Feathers may have become increasingly aerodynamic over millions of years, eventually allowing dinosaurs to glide from tree to tree. Individuals able to perform such
90 a feat might have been able to reach new food sources or better escape predators—and pass the trait on to subsequent generations.

31. The overall structure of the passage is best described as an account of a unique fossil discovery followed by:
- A. an analysis of how the tests typically performed on fossils have changed since the discovery.
 - B. a description of what paleontologists believed about dinosaurs before the fossil discovery.
 - C. an overview of theories that may help answer a question raised by the fossil.
 - D. an explanation of the process by which fossils are identified and classified.

32. In the context of the passage, the phrase “turn paleontology on its head” (lines 17–18) most nearly refers to the way that Li’s fossil discovery would cause paleontologists to:
- F. revise their understanding of the evolutionary links between dinosaurs and birds.
 - G. allow fossils unearthed by amateur fossil hunters to be formally studied.
 - H. adopt new, more collaborative processes for identifying fossils.
 - J. debate whether birds should be classified as prehistoric.
33. The passage suggests that one reason fossils of feathered dinosaurs are significant is that they may offer information about:
- A. whether other ancient species had feather-like structures.
 - B. whether feathers have a relationship to filaments.
 - C. why some modern birds do not fly.
 - D. how feathers evolved to aid flight.
34. In the context of the passage, the information about Xu in lines 43–47 primarily serves to:
- F. provide more details about the discoverer of *Sinosauropteryx*.
 - G. introduce the scientist who developed the theories about feathered dinosaurs presented in the rest of the passage.
 - H. indicate that the discovery of ancient feathers has led to the creation of a specialized institute.
 - J. lend credibility to the theories about feather evolution presented in the following paragraph.
35. The idea that feathers helped dinosaurs scramble up tree trunks is most nearly presented in the passage as:
- A. a fact established by the presence of feather-like structures in *Sinosauropteryx*.
 - B. a reasoned judgment supported by the observation that animals descended from dinosaurs perform a similar feat.
 - C. an opinion referenced by Carrano to show how speculative some studies of feathered dinosaurs have been.
 - D. an opinion held by Carrano despite Zhang’s findings to the contrary.
36. The main idea of the second paragraph (lines 10–14) is that Li’s fossil was:
- F. displayed to the public as a bird despite research proving it to be a dinosaur.
 - G. removed from a museum to undergo further study by paleontologists.
 - H. believed to be a bird by some experts and a dinosaur by others.
 - J. ignored by paleontologists but valued by a museum director.
37. The passage indicates that many feathered dinosaurs could not have used feathers to:
- A. achieve flight.
 - B. repel water.
 - C. climb trees.
 - D. retain heat.
38. As it is used in line 42, the word *forged* most nearly means:
- F. accelerated.
 - G. mimicked.
 - H. falsified.
 - J. shaped.
39. According to the passage, dinosaurs typically couldn’t climb well because:
- A. they had filaments instead of feathers.
 - B. their limbs were too short.
 - C. they had hinged ankles.
 - D. they were too heavy.
40. The passage author claims that being better able to escape predators and reach new food sources may have been instrumental in helping feathered dinosaurs:
- F. pass the trait of feathers on to new generations.
 - G. develop the muscles required to fly.
 - H. attract a wider variety of mates.
 - J. shed their quill-like filaments.

END OF TEST 3

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.

DO NOT RETURN TO A PREVIOUS TEST.



SCIENCE TEST

35 Minutes—40 Questions

DIRECTIONS: There are several passages in this test. Each passage is followed by several questions. After reading a passage, choose the best answer to each question and fill in the corresponding oval on your answer document. You may refer to the passages as often as necessary.

You are NOT permitted to use a calculator on this test.

Passage I

On Jupiter's moon Europa, liquid water may be present beneath the outer layer of frozen water. Scientists investigated this possibility using 2 computer models (A and B). Both models determined the effects of the impact of each of 2 comets—a 0.5 km diameter comet and a 1.0 km diameter comet. In Model A, each comet struck a 6 km layer of frozen water. In Model B, each comet struck a 5 km layer of frozen water over a 1 km layer of liquid water. In each model, the comets had the same composition, a density of 1.1 g/cm^3 , an impact angle of 90° (with the surface), and an impact speed of 26.5 km/sec .

Figures 1 and 2 show the effects.

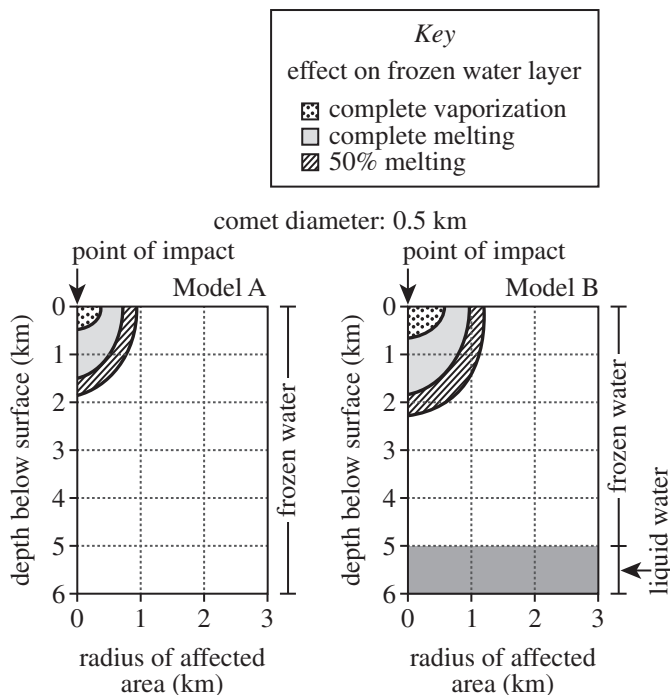


Figure 1

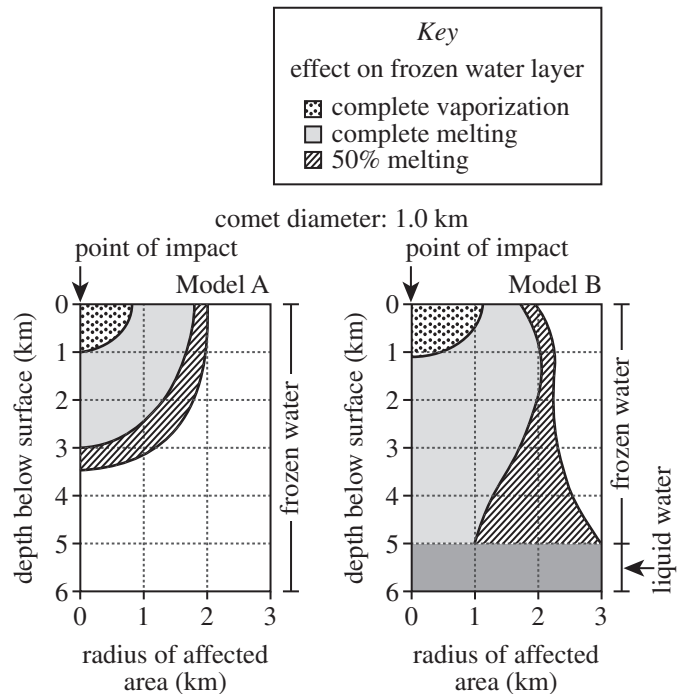


Figure 2

Figures adapted from E. P. Turtle and E. Pierazzo, "Thickness of a European Ice Shell from Impact Simulations." ©2001 by the American Association for the Advancement of Science.



1. According to the results of Model A in Figure 1, the maximum depth at which there is 50% melting is closest to which of the following?
 - A. 1.5 km
 - B. 1.8 km
 - C. 2.1 km
 - D. 2.4 km
2. According to the results of Model B in Figure 2, at the surface, the maximum radius at which there is complete vaporization is closest to which of the following?
 - F. 0.8 km
 - G. 1.1 km
 - H. 1.4 km
 - J. 1.7 km
3. According to the results of which model in Figure 1 is a greater area of the surface affected by the impact, and according to the results of which model in Figure 1 does the impact have a deeper effect?

	greater area of surface affected	deeper effect
A.	Model A	Model A
B.	Model A	Model B
C.	Model B	Model A
D.	Model B	Model B
4. Based on the results of Model B in Figure 2, the H₂O that is present at a radius of 2 km and a depth of 4 km exists in what state(s)?
 - F. Solid only
 - G. Liquid only
 - H. Solid and liquid only
 - J. Solid and gas only
5. According to the results of Model B in Figure 2, from a depth of 3 km to a depth of 5 km, does the maximum radius at which there is complete melting decrease or increase, and does the maximum radius at which there is 50% melting decrease or increase?

	complete melting	50% melting
A.	decrease	decrease
B.	decrease	increase
C.	increase	decrease
D.	increase	increase
6. Consider the results of Model A in Figure 2. Those results indicate that within an area having a radius of 0.5 km, the material at the surface changes from a:
 - F. gas to a liquid.
 - G. liquid to a solid.
 - H. solid to a gas.
 - J. gas to a solid.

**Passage II**

N. crassa, a fungus, cannot grow without the amino acid *arginine*. Normal *N. crassa* absorb arginine from the environment or synthesize it. Part of the biochemical pathway for the synthesis of arginine is shown in Figure 1.

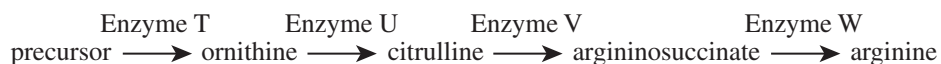


Figure 1

Three mutant strains of *N. crassa* (Strains 1–3), each with a defect in one of the 4 enzymes listed in Figure 1, were identified. Each strain was tested for the ability to grow on 4 mediums: GM (which contained the precursor but lacked ornithine, citrulline, argininosuccinate, and arginine); GM + ornithine; GM + citrulline; and GM + arginine. A normal strain of *N. crassa* (Strain 4) was also tested. The results are shown in Table 1.

Table 1				
Strain	Growth on:			
	GM	GM + ornithine	GM + citrulline	GM + arginine
1	no	yes	yes	yes
2	no	no	yes	yes
3	no	no	no	yes
4	yes	yes	yes	yes

Figure 1 and Table 1 adapted from Jane B. Reece et al., *Campbell Biology*. ©2011 by Pearson Education, Inc.



7. Based on Figure 1 and Table 1, which of the strains most likely had a defect in either Enzyme V or Enzyme W?
- Strain 1
 - Strain 2
 - Strain 3
 - Strain 4
8. Based on Figure 1, Enzyme U facilitates the synthesis of arginine by:
- speeding up the reaction that converts ornithine to citrulline.
 - speeding up the reaction that converts citrulline to ornithine.
 - slowing down the reaction that converts ornithine to citrulline.
 - slowing down the reaction that converts citrulline to ornithine.
9. Strains 1–3 could not grow on GM because they could not synthesize arginine, which is a subunit of:
- carbohydrates.
 - proteins.
 - DNA.
 - RNA.
10. Based on Figure 1 and Table 1, which of Strains 1 and 2, if either, would have been able to grow on GM + argininosuccinate?
- Strain 1 only
 - Strain 2 only
 - Both Strain 1 and Strain 2
 - Neither Strain 1 nor Strain 2
11. A scientist hypothesized that a newly identified mutant strain of *N. crassa* (Strain 5) had a defect in the same enzyme as Strain 3. Strain 5 was tested according to the procedure described in the passage, and the results of the test are shown below. Are these results consistent with the scientist's hypothesis?
- | Strain | Growth on: | | | |
|--------|------------|----------------|-----------------|---------------|
| | GM | GM + ornithine | GM + citrulline | GM + arginine |
| 5 | no | no | yes | yes |
- Yes, because Strain 5 could grow only on GM and GM + ornithine.
 - Yes, because Strain 5 could grow only on GM + citrulline and GM + arginine.
 - No, because Strain 5 could grow on both GM and GM + ornithine.
 - No, because Strain 5 could grow on both GM + citrulline and GM + arginine.
12. How many of the strains, if any, could convert the precursor to arginine?
- 0
 - 1
 - 3
 - 4

**Passage III**

Environmental conditions, such as nighttime temperature, can affect the reproduction of *Uta stansburiana* (a species of lizard). A scientist conducted a study to examine how nighttime temperature affects egg laying by female *U. stansburiana*.

Study

Each of 20 identical tanks with 25 cm high glass walls was prepared as follows: First, sand was poured into the tank to a depth of 10 cm. Next, three 1-year-old female *U. stansburiana* were artificially fertilized and then placed into the tank. Finally, the top of the tank was covered with a fine wire mesh. After the tanks were prepared, they were equally divided into 4 groups (Groups Q–T).

Group Q was incubated in the presence of fluorescent light for 14 hr at 29°C. Then Group Q was incubated in the dark for 10 hr at 8°C. This procedure was repeated every day for the next 39 days. During the incubation period, any eggs laid were collected, and the *average clutch size* (average number of eggs laid per egg-laying female), the *average egg mass* (average mass per egg), and the *average clutch time* (average time required by a female to lay a clutch of eggs) were determined for Group Q.

The procedure for Group Q was then repeated for Groups R–T, except that instead of being incubated in the dark at 8°C, Group R was incubated in the dark at 18°C, Group S was incubated in the dark at 22°C, and Group T was incubated in the dark at 29°C.

The results for the 4 groups are shown in the following table.

Group	Average clutch size (eggs)	Average egg mass (mg)	Average clutch time (days)
Q	3.8	300	35.5
R	4.5	310	32.9
S	4.8	290	31.0
T	4.1	330	26.5

Table adapted from Donald N. Clarke and Peter A. Zani, "Effects of Night-Time Warming on Temperate Ectotherm Reproduction: Potential Fitness Benefits of Climate Change for Side-Blotched Lizards." ©2012 by The Company of Biologists Ltd.

13. What percent of the height of a tank was filled with sand?

- A. 20%
- B. 40%
- C. 60%
- D. 80%

14. Which of the statements listed below describe(s) a function of the fine wire mesh?

- I. To place the *U. stansburiana* in a tank
- II. To protect the *U. stansburiana* from attack by prey
- III. To prevent the *U. stansburiana* from leaving a tank

- F. I only
- G. III only
- H. I and II only
- J. I, II, and III

15. Assume that the greater an egg's mass, the healthier its embryo. Based on the results of the study, among the 4 temperatures at which the *U. stansburiana* were incubated in the dark, the temperature that most likely resulted in the healthiest embryos, on average, was:

- A. 8°C.
- B. 18°C.
- C. 22°C.
- D. 29°C.

16. Is *U. stansburiana* a reptile or a mammal, and does *U. stansburiana* primarily maintain its body temperature by generating heat from internal metabolic processes or by absorbing heat from the surrounding environment?

- F. Reptile; absorbing heat from the surrounding environment
- G. Reptile; generating heat from internal metabolic processes
- H. Mammal; absorbing heat from the surrounding environment
- J. Mammal; generating heat from internal metabolic processes



17. Consider the steps listed below that were performed in the study.

1. Place the *U. stansburiana* in a tank.
2. Place sand in a tank.
3. Place wire mesh over the top of a tank.
4. Artificially fertilize the *U. stansburiana*.

These steps were performed in what order?

- A. 1, 2, 3, 4
- B. 1, 4, 3, 2
- C. 2, 1, 4, 3
- D. 2, 4, 1, 3

18. For which group was the incubation temperature constant over a 24 hr period?

- F. Group Q
- G. Group R
- H. Group S
- J. Group T

19. The incubation conditions for Group R and Group T differed in which of the following ways? The incubation temperature in the:

- A. absence of light was higher for Group R than for Group T.
- B. absence of light was higher for Group T than for Group R.
- C. presence of light was higher for Group R than for Group T.
- D. presence of light was higher for Group T than for Group R.



Passage IV

Consider a pendulum of length L that consists of a sphere of mass M suspended from one end of a string of negligible mass; the string's other end is tied to a ring stand (see Figure 1).

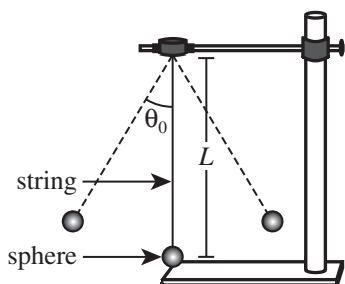


Figure 1

When the sphere is moved from its *equilibrium* (lowest) position, the string makes some initial angle, θ_0 , with respect to the vertical direction. If the sphere is released, the pendulum will *oscillate* (swing back and forth). The frequency, f , of the pendulum is the number of *cycles* (round trips) per second completed by the pendulum.

A teacher gave her students the following table of masses and lengths for 4 such pendulums, P1–P4.

Table 1		
Pendulum	M (kg)	L (m)
P1	1	1
P2	1	2
P3	2	1
P4	2	2

She asked 2 students to predict which pendulum(s) would have the greatest f (assuming that θ_0 is the same for the 4 pendulums).

Student 1

For a given L , as M increases, the gravitational force exerted by Earth on a sphere will increase, so sphere weight will increase. As this weight increases, gravitational acceleration will increase, so the average speed of the sphere over a cycle will increase, thereby increasing f .

For a given M , as L decreases, the distance traveled by the sphere over a cycle will decrease. However, the average sphere speed will also decrease, because the length of the path traveled by the sphere over a cycle will decrease and limit gravitational acceleration. Consequently, the pendulum will take the same amount of time to complete a cycle, and f will remain the same.

Because P3 and P4 have the same M , they will have the same f . Moreover, because both P3 and P4 have a greater mass than either P1 or P2, each will have a greater f than either P1 or P2.

Student 2

M will not affect f , because a sphere's gravitational acceleration and, consequently, its average speed over a cycle are independent of the sphere's mass.

For any given M , as L increases, the distance traveled by the sphere over a cycle will increase. As a result, the pendulum will take more time to complete a cycle, so f will decrease.

As P1 and P3 have the same L , they will have the same f . As both P1 and P3 are shorter than either P2 or P4, each will have a greater f than either P2 or P4.



20. Based on Table 1 and Student 1's predictions, which pendulum will have the greater gravitational acceleration, P1 or P3 ?
- F. P1, because it has the greater mass.
 - G. P1, because it has the lesser mass.
 - H. P3, because it has the greater mass.
 - J. P3, because it has the lesser mass.
21. Suppose that the spheres are removed from P2 and P4, and then the spheres are simultaneously dropped from a height of 5 m above the ground. Based on Table 1 and Student 1's predictions, the sphere more likely to strike the ground first will be the sphere removed from:
- A. P2, because its average speed will be greater.
 - B. P2, because its average speed will be less.
 - C. P4, because its average speed will be greater.
 - D. P4, because its average speed will be less.
22. Two pendulums, P5 and P6, have the same f . How would Student 2 most likely explain this result?
- F. P5 and P6 have the same L .
 - G. P5 and P6 do not have the same L .
 - H. P5 and P6 have the same M .
 - J. P5 and P6 do not have the same M .
23. Based on Table 1 and Student 2's predictions, which pendulum will travel the greater distance over a cycle, P3 or P4 ?
- A. P3, because it has the greater L .
 - B. P3, because it has the lesser L .
 - C. P4, because it has the greater L .
 - D. P4, because it has the lesser L .
24. Assume that if $M = 1.0$ kg and $L = 1.0$ m, f will equal 0.50 cycle per sec. Based on Student 1's predictions, if $M = 2.0$ kg and $L = 1.0$ m, f will most likely be:
- F. greater than 0.50 cycle per sec.
 - G. equal to 0.50 cycle per sec.
 - H. greater than 0.0 cycle per sec but less than 0.50 cycle per sec.
 - J. equal to 0.0 cycle per sec.
25. Consider the statements about f , M , and L given below.
- I. The frequency f is affected by M but not by L .
 - II. The frequency f is affected by L but not by M .
 - III. The frequency f is affected by both M and L .
 - IV. The frequency f is affected by neither M nor L .
- Both students would most likely *disagree* with which of the statements?
- A. I only
 - B. III only
 - C. I and II only
 - D. III and IV only
26. Which of the 2 students, if either, would be likely to agree that as sphere mass increases, frequency will decrease?
- F. Student 1 only
 - G. Student 2 only
 - H. Both students
 - J. Neither student



Passage V

Coffee beans slowly release CO_2 gas after they have been roasted. Two experiments were done to compare the amounts of CO_2 released by various samples of roasted coffee beans.

In each trial of the 2 experiments, Steps 1–3 were performed:

1. A 250.0 g sample of freshly roasted coffee beans (either whole or ground) was poured through the inlet port of the apparatus shown below.

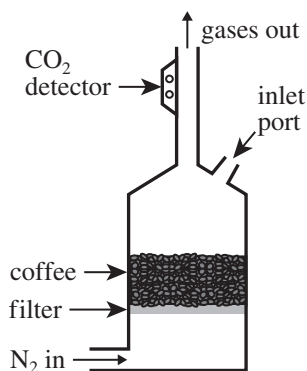


diagram of apparatus

2. The inlet port was sealed, and a flow of N_2 gas was forced through the apparatus for 12 hr. None of the N_2 was absorbed by the beans.
3. Over the 12 hr period, the CO_2 detector measured the total amount of CO_2 that was released by the beans.

Experiment 1

Each of six 250.0 g samples of whole coffee beans was roasted according to a different set of time and temperature conditions. The 6 sets of conditions, each of which resulted in a particular roast color, are shown in Table 1.

Roasting time	Roasting temperature ($^{\circ}\text{C}$)	Roast color
short (240 sec)	230	light-S
	240	medium-S
	250	dark-S
long (600 sec)	200	light-L
	210	medium-L
	220	dark-L

After a sample of beans was roasted, Steps 1–3 were performed. Figure 1 shows the total amount of CO_2 released by each sample (in milligrams of CO_2 released per gram of beans, mg/g).

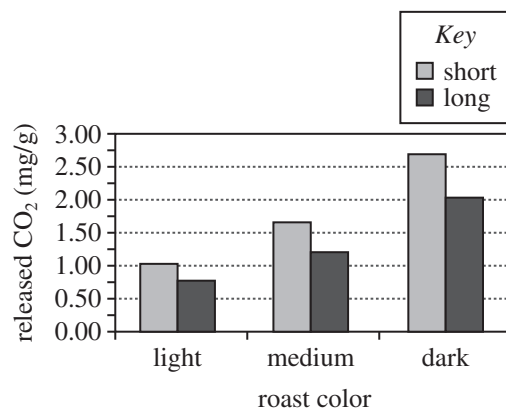


Figure 1

Experiment 2

Three groups of whole coffee beans were roasted for 240 sec such that each acquired a different roast color. Next, each group of beans was ground into 3 batches, each having a different average particle size. Then a 250.0 g sample of each batch was collected. After a sample was collected, Steps 1–3 were performed. Figure 2 shows the total amount of CO_2 released by each of the 9 samples.

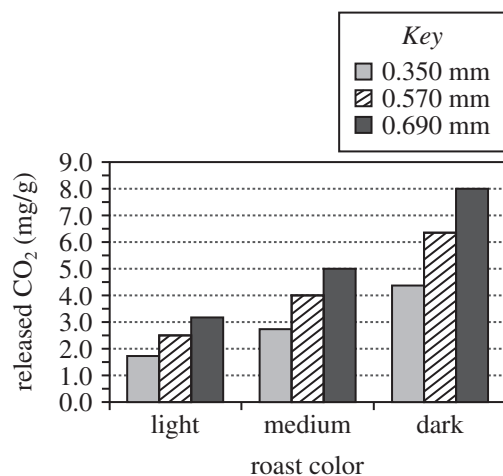


Figure 2

Table and figures adapted from B. A. Anderson et al., "The Diffusion Kinetics of Carbon Dioxide in Fresh Roasted and Ground Coffee." ©2003 by Elsevier Science Ltd.



27. Consider the statement “As the average particle size increased, the total amount of released CO_2 increased.” This statement is supported by the results of Experiment 2 for which of the roast colors?
- A. Light and medium only
 - B. Light and dark only
 - C. Medium and dark only
 - D. Light, medium, and dark
28. In any given trial of the experiments, if the N_2 gas flowed through the apparatus at a rate of 2.0 L/hr, how many liters of N_2 flowed through the apparatus?
- F. 6 L
 - G. 24 L
 - H. 48 L
 - J. 72 L
29. In Experiment 1, for a roasting time of 240 sec, as roasting temperature increased, the amount of released CO_2 :
- A. increased only.
 - B. decreased only.
 - C. increased and then decreased.
 - D. decreased and then increased.
30. Suppose that a trial had been performed in Experiment 2 with a sample of light-roast beans ground to an average particle size of 0.420 mm. The total amount of CO_2 released by the sample would most likely have been:
- F. less than 1.7 mg/g.
 - G. between 1.7 mg/g and 2.5 mg/g.
 - H. between 2.5 mg/g and 3.2 mg/g.
 - J. greater than 3.2 mg/g.
31. Suppose that, in the trial of Experiment 1 involving the medium-S sample, the inlet port had NOT been sealed after the sample was poured into the apparatus. Would the total amount of CO_2 measured by the detector during this trial more likely have been greater than 1.6 mg/g or less than 1.6 mg/g ?
- A. Greater, because CO_2 from the beans would have escaped from the apparatus through the inlet port.
 - B. Greater, because CO_2 from the air would have entered the apparatus through the inlet port.
 - C. Less, because CO_2 from the beans would have escaped from the apparatus through the inlet port.
 - D. Less, because CO_2 from the air would have entered the apparatus through the inlet port.
32. A scientist predicted that, on average, all the coffee beans roasted at a temperature greater than 235°C would release more than 1.50 mg of CO_2 /g. Do the results of Experiment 1 support this prediction?
- F. Yes; both the medium-S and dark-S samples released more than 1.50 mg of CO_2 /g.
 - G. Yes; both the medium-L and dark-L samples released more than 1.50 mg of CO_2 /g.
 - H. No; both the medium-S and dark-S samples released less than 1.50 mg of CO_2 /g.
 - J. No; both the medium-L and dark-L samples released less than 1.50 mg of CO_2 /g.
33. Upon the completion of Experiment 1, what was the minimum number of hours that N_2 gas had flowed through the apparatus?
- A. 36 hr
 - B. 48 hr
 - C. 72 hr
 - D. 96 hr



Passage VI

Carotenoids (CTs) are colored compounds that are found in many foods. Solutions of CTs can absorb ultra-violet (UV) light, which has wavelengths ranging from 200 nanometers (nm) to 400 nm, and visible light, which has wavelengths ranging from 400 nm to 700 nm. (1 nm = 10^{-9} m.) The amount of light absorbed (the *absorbance*, A) by a CT solution varies with wavelength, λ . The wavelength that results in the maximum A is known as $\lambda_{\max}$.

When a CT solution absorbs visible light of a certain color at $\lambda_{\max}$, the solution appears to be a different color. On a color wheel (Figure 1), the color of the solution is across from the color of light absorbed.

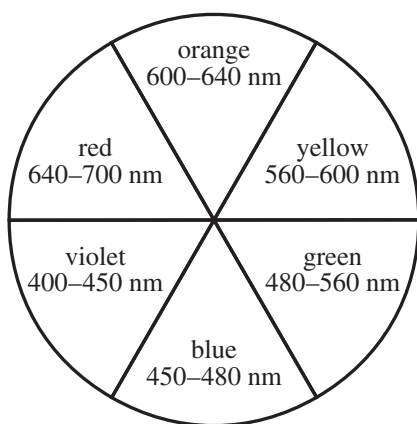


Figure 1

For example, if a CT solution has a $\lambda_{\max}$ of 642 nm, then the solution absorbs red light and appears to be green.

Study

Students determined the $\lambda_{\max}$ of each of 6 CTs in solution. For each CT, Steps 1–5 were performed:

1. The CT was dissolved in a certain solvent at 25°C to make 5 mL of solution.
2. A 1 mL sample of the CT solution was placed in a plastic vial, and this vial was placed in a *spectrophotometer* (a device that measures how a solution's A varies with λ).

3. Light from 200 nm through 700 nm was directed through the sample onto a detector (see Figure 2). Some of the light was absorbed by the sample and thus did not reach the detector.

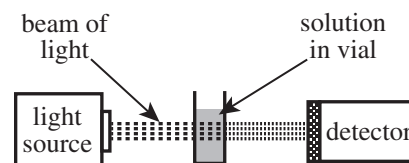


Figure 2

4. The sample's A was determined at every 1 nm.
5. The $\lambda_{\max}$ was identified.

Table 1 lists the CTs and the $\lambda_{\max}$ for each.

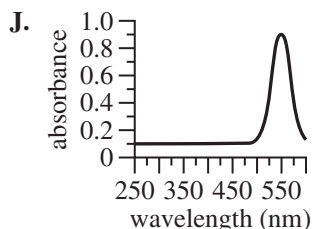
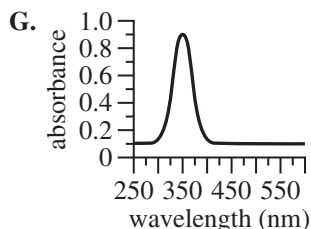
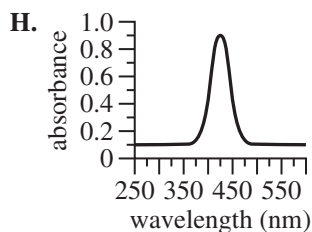
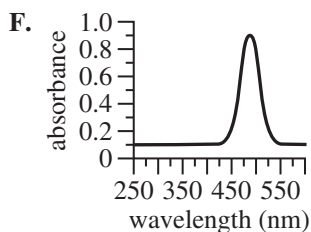
Table 1	
CT	$\lambda_{\max}$ (nm)
Phytoene	286
Auroxanthin	399
Beta-carotene	451
Lycopene	468
Astaxanthin	488
Spirilloxanthin	492

Table 1 adapted from G. Britton, S. Liaaen-Jensen, and H. Pfander, eds., "Carotenoids, Volume 1B: Spectroscopy." ©1995 by Birkhäuser Verlag.

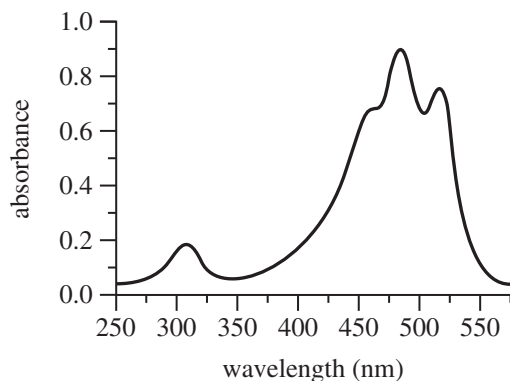
34. The solution containing which of the following CTs absorbed green light at its $\lambda_{\max}$?
 - F. Auroxanthin
 - G. Beta-carotene
 - H. Lycopene
 - J. Astaxanthin
35. What percent of the volume of the CT solution prepared in Step 1 was placed into a vial in Step 2 ?
 - A. 10%
 - B. 20%
 - C. 50%
 - D. 80%



36. Which of the following graphs best shows how the absorbance of the astaxanthin sample varied with wavelength?



37. Consider the following results for a CT solution of *torulene*.



At its λ_{max} , does the torulene solution absorb UV light or visible light?

- A. UV light, because λ_{max} is between 200 nm and 400 nm.
 B. UV light, because λ_{max} is between 400 nm and 700 nm.
 C. Visible light, because λ_{max} is between 200 nm and 400 nm.
 D. Visible light, because λ_{max} is between 400 nm and 700 nm.

38. Consider the vial that contained each CT solution when in the spectrophotometer. Based on the description of the study and Figure 2, is it more likely that the students chose to use this vial because it was composed of a plastic that transmitted all or almost all light between 200 nm and 700 nm or because it was composed of a plastic that reflected all or almost all light between 200 nm and 700 nm?

- F. Transmitted, so that the light would pass into the solution contained in the vial.
 G. Transmitted, so that the light would be prevented from passing into the solution contained in the vial.
 H. Reflected, so that the light would pass into the solution contained in the vial.
 J. Reflected, so that the light would be prevented from passing into the solution contained in the vial.

39. *Violerythrin* is a CT that appears violet when in solution. Based on Figure 1, the λ_{max} of a violerythrin solution could be which of the following values?

- A. 513 nm
 B. 566 nm
 C. 643 nm
 D. 698 nm

40. Each time a sample's A was determined at a specific λ in Step 4, the sample is said to have been *scanned*. Based on the description of the study, the total number of scans performed in the study was approximately:

- F. 200.
 G. 300.
 H. 1,200.
 J. 3,000.

END OF TEST 4

STOP! DO NOT RETURN TO ANY OTHER TEST.

Scoring Guide

Raw Scores

The number of questions you answered correctly on each test section is a raw score. Because there are many forms of the ACT, each with different questions, the difficulty level varies between the forms. A raw score of 35 on one form of the mathematics test section, for example, may be about as difficult to earn as a raw score of 37 on another form of that test section.

Computing raw scores: To compute your raw scores, check your answers with the scoring information in the scoring keys and conversion table and do the following:

1. Mark a one (1) in the blank for each question answered correctly.
2. Count the number of correct answers for each of the four multiple-choice test sections.
3. Add up the total number correct for each category and test section and capture it as directed above its scoring key.

These numbers are your raw scores on the individual multiple-choice test sections. The highest raw score for a given test section is the number of questions included on that test section:

- English: 75
- Mathematics: 60
- Reading: 40
- Science: 40

Note: Raw scores are also provided as part of the Summary Report for each test section taken online in TestNav.

English Test 1 Section

English Scoring Key (for Form Z14)

English Reporting Categories
(Capture raw scores/correct answers.)

Production of Writing (POW) = ____ of 23

Knowledge of Language (KLA) = ____ of 12

Conventions of Standard English (CSE) = ____ of 40

Total English (POW + KLA + CSE) = ____ of 75

English Number	Correct Answer	Correct (Mark 1)	Reporting Categories
1	D		CSE
2	G		CSE
3	A		CSE
4	J		POW
5	A		CSE
6	G		CSE
7	C		CSE
8	F		CSE
9	C		CSE
10	J		CSE
11	B		POW
12	H		CSE
13	D		KLA
14	G		POW
15	B		POW
16	F		CSE
17	C		CSE
18	J		CSE
19	C		CSE
20	G		POW
21	D		KLA
22	G		CSE
23	D		POW
24	F		CSE
25	B		CSE

English Number	Correct Answer	Correct (Mark 1)	Reporting Categories
26	F		POW
27	D		CSE
28	J		KLA
29	A		CSE
30	H		POW
31	C		CSE
32	F		CSE
33	A		CSE
34	F		CSE
35	C		KLA
36	H		POW
37	D		KLA
38	F		CSE
39	B		POW
40	G		POW
41	A		CSE
42	J		POW
43	B		CSE
44	H		POW
45	B		CSE
46	H		POW
47	B		CSE
48	H		POW
49	D		CSE
50	F		KLA

English Number	Correct Answer	Correct (Mark 1)	Reporting Categories
51	B		CSE
52	J		CSE
53	A		KLA
54	F		CSE
55	D		CSE
56	G		POW
57	D		KLA
58	J		CSE
59	A		POW
60	H		POW
61	C		CSE
62	F		CSE
63	C		KLA
64	J		CSE
65	D		KLA
66	G		POW
67	C		CSE
68	J		POW
69	A		KLA
70	H		CSE
71	B		KLA
72	F		POW
73	C		CSE
74	H		POW
75	B		POW

Mathematics Test 2 Section

Mathematics Scoring Key (for Form Z14)

Mathematics Reporting Categories

(Capture raw scores/correct answers.)

Preparing for Higher Math (PHM) = ___ of 35

(A + F + G + N + S)

A = Algebra

F = Functions

G = Geometry

N = Number & Quantity

S = Statistics & Probability

Integrating Essential Skills (IES) = ___ of 25

Modeling (MDL) = ___ of 24

Total Mathematics (PHM + IES) = ___ of 60

Note: Do not include MDL in Total Mathematics raw score.

Math Number	Correct Answer	Correct (Mark 1)	Reporting Categories
1	C		PHM-G
2	F		PHM-F
3	A		PHM-A
4	K		IES, MDL
5	D		IES
6	F		IES, MDL
7	A		PHM-G
8	G		IES, MDL
9	E		PHM-F
10	H		PHM-S, MDL
11	C		IES, MDL
12	H		PHM-A
13	A		PHM-G, MDL
14	J		IES
15	B		IES, MDL
16	F		PHM-G, MDL
17	E		PHM-A
18	F		IES
19	B		IES
20	H		PHM-A
21	D		PHM-A

Math Number	Correct Answer	Correct (Mark 1)	Reporting Categories
22	J		PHM-N
23	C		PHM-S, MDL
24	F		PHM-N
25	C		IES
26	G		PHM-F
27	E		IES
28	J		IES, MDL
29	B		IES, MDL
30	H		PHM-G, MDL
31	D		IES, MDL
32	G		IES, MDL
33	C		PHM-N
34	G		IES
35	D		PHM-S, MDL
36	K		IES, MDL
37	D		PHM-G
38	J		PHM-F
39	B		PHM-F
40	K		PHM-A
41	E		IES, MDL
42	K		IES, MDL
43	E		PHM-A
44	F		PHM-S, MDL
45	B		IES, MDL
46	H		PHM-F
47	C		PHM-N, MDL
48	G		IES
49	B		PHM-F
50	F		PHM-F
51	D		PHM-N
52	G		PHM-S, MDL
53	D		PHM-A
54	K		IES
55	E		IES
56	K		IES
57	A		PHM-G
58	K		PHM-G
59	A		PHM-S, MDL
60	J		IES, MDL

Reading Test 3 Section

Reading Scoring Key (for Form Z14)

Reading Reporting Categories

(Capture raw scores/correct answers.)

Key Ideas & Details (KID) = ___ of 23

Craft & Structure (CS) = ___ of 11

Integration of Knowledge & Ideas (IKI) = ___ of 6

Total Reading (KID + CS + IKI) = ___ of 40

Reading Number	Correct Answer	Correct (Mark 1)	Reporting Categories
1	A		KID
2	H		CS
3	A		KID
4	G		KID
5	D		KID
6	G		KID
7	A		CS
8	J		KID
9	B		KID
10	H		IKI
11	D		KID
12	G		KID
13	C		CS
14	H		KID

Reading Number	Correct Answer	Correct (Mark 1)	Reporting Categories
15	D		CS
16	F		CS
17	D		KID
18	J		IKI
19	C		IKI
20	F		IKI
21	B		KID
22	G		CS
23	D		KID
24	F		KID
25	A		KID
26	G		KID
27	A		KID
28	J		CS
29	D		KID
30	G		KID
31	C		CS
32	F		CS
33	D		KID
34	J		CS
35	B		IKI
36	H		KID
37	A		KID
38	J		CS
39	C		KID
40	F		IKI

Science Test 4 Section

Science Scoring Key (for Form Z14)

Science Reporting Categories

(Capture raw scores/correct answers.)

Interpretation of Data (IOD) = ____ of 17

Scientific Investigation (SIN)= ____ of 12

Evaluation of Models, Inferences
& Experimental Results (EMI) = ____ of 11

Total Science (IOD + SIN + EMI) = ____ of 40

Science Number	Correct Answer	Correct (Mark 1)	Reporting Categories
1	B		IOD
2	G		IOD
3	D		IOD
4	H		IOD
5	B		IOD
6	H		IOD
7	C		IOD
8	F		IOD
9	B		IOD
10	H		IOD
11	D		EMI
12	G		IOD
13	B		SIN
14	G		SIN

Science Number	Correct Answer	Correct (Mark 1)	Reporting Categories
15	D		EMI
16	F		IOD
17	D		SIN
18	J		SIN
19	B		SIN
20	H		EMI
21	C		EMI
22	F		EMI
23	C		EMI
24	F		EMI
25	D		EMI
26	J		EMI
27	D		EMI
28	G		SIN
29	A		IOD
30	G		SIN
31	C		SIN
32	F		EMI
33	C		SIN
34	J		IOD
35	B		SIN
36	F		IOD
37	D		IOD
38	F		SIN
39	B		IOD
40	J		SIN

Scale Scores

To adjust for the small differences among different forms of the ACT test, raw scores are converted into scale scores. Scale scores appear on reports sent to your school.

When your raw scores are converted into scale scores, it becomes possible to compare your scores with those of examinees who took different test forms. For example, a scale score of 26 on the mathematics test section has the same meaning for any form of the ACT.

Converting Raw Scores to Scale Scores

Each ACT test section generates a single scale score of 1–36. Use the Conversion of Raw Scores to Scale Scores table to convert your raw scores to scale scores. For each test section:

1. Locate your raw score, or the range of raw scores that includes it, in the conversion table.
2. Locate the cell in the Scale Score column of the row that corresponds to that raw score to identify the scale score for that test section.
3. As you determine your scale scores, enter them in the blanks below.

Test section scale scores:

English = ____
Mathematics = ____
Reading = ____
Science = ____

Calculating a Composite Score

An ACT test generates a single Composite score of 1–36. Compute the Composite score by averaging the four scale scores. To do this:

1. Add your four scale scores. Enter this sum of scores in the blank below.
2. Divide the sum by 4. If the resulting number ends in a fraction, round it to the nearest whole number. (Round down any fraction less than one-half, except for averages lower than one; round up any fraction that is one-half or more and/or less than one.)
3. Enter this number in the blank below. This is your Composite score.

Composite of scale scores:

Sum of scores = ____
Composite score (sum ÷ 4) = ____

Note: If you left a test section completely blank and marked no items, do not list a scale score for that section and do not calculate a Composite score.

Conversion of Raw Scores to Scale Scores (for Form Z14)

Scale Score	Raw Score Test 1: English	Raw Score Test 2: Mathematics	Raw Score Test 3: Reading	Raw Score Test 4: Science
36	73–75	58–60	39–40	39–40
35	70–72	55–57	38	38
34	68–69	53–54	37	37
33	67	52	36	36
32	66	50–51	35	35
31	65	49	34	—
30	64	47–48	—	34
29	63	45–46	33	33
28	62	43–44	32	32
27	60–61	40–42	31	31
26	59	38–39	30	29–30
25	56–58	36–37	29	28
24	54–55	34–35	28	26–27
23	51–53	33	26–27	24–25
22	48–50	31–32	24–25	22–23
21	45–47	30	23	21
20	43–44	28–29	21–22	19–20
19	41–42	26–27	20	18
18	39–40	24–25	18–19	16–17
17	37–38	21–23	17	14–15
16	34–36	17–20	16	13
15	31–33	14–16	14–15	12
14	28–30	10–13	12–13	11
13	26–27	7–9	11	10
12	24–25	6	9–10	9
11	21–23	5	8	8
10	18–20	4	7	6–7
9	16–17	3	6	5
8	13–15	—	5	—
7	11–12	2	4	4
6	9–10	—	—	3
5	7–8	—	3	2
4	5–6	1	2	—
3	4	—	—	1
2	2–3	—	1	—
1	0–1	0	0	0



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